Introduction to Aerospace Engineering (R18A2101)

COURSE FILE

II B. Tech I Semester

(2019-2020)

Prepared By

Prof Vedantam Ravi

Department of Aeronautical Engineering

MALLA REDDY COLLEGE OF ENGINEERING & TECHNOLOGY

(Autonomous Institution – UGC, Govt. of India)

Affiliated to JNTU, Hyderabad, Approved by AICTE - Accredited by NBA & NAAC – 'A' Grade - ISO 9001:2015 Certified)

Maisammaguda, Dhulapally (Post Via. Kompally), Secunderabad – 500100, Telangana State, India.

MRCET VISION

- To become a model institution in the fields of Engineering, Technology and Management.
- To have a perfect synchronization of the ideologies of MRCET with challenging demands of International Pioneering Organizations.

MRCET MISSION

To establish a pedestal for the integral innovation, team spirit, originality and competence in the students, expose them to face the global challenges and become pioneers of Indian vision of modern society.

MRCET QUALITY POLICY.

- To pursue continual improvement of teaching learning process of Undergraduate and Post Graduate programs in Engineering & Management vigorously.
- To provide state of art infrastructure and expertise to impart the quality education.

PROGRAM OUTCOMES
(PO's)

Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design / development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multi disciplinary environments.
12. **Life- long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

DEPARTMENT OF AERONAUTICAL ENGINEERING

VISION

Department of Aeronautical Engineering aims to be indispensable source in Aeronautical Engineering which has a zeal to provide the value driven platform for the students to acquire knowledge and empower themselves to shoulder higher responsibility in building a strong nation.

MISSION

The primary mission of the department is to promote engineering education and research. To strive consistently to provide quality education, keeping in pace with time and technology. Department passions to integrate the intellectual, spiritual, ethical and social development of the students for shaping them into dynamic engineers.

QUALITY POLICY STATEMENT

Impart up-to-date knowledge to the students in Aeronautical area to make them quality engineers. Make the students experience the applications on quality equipment and tools. Provide systems, resources and training opportunities to achieve continuous improvement. Maintain global standards in education, training and services.

PROGRAM EDUCATIONAL OBJECTIVES – Aeronautical Engineering

1. **PEO1 (PROFESSIONALISM & CITIZENSHIP):** To create and sustain a community of learning in which students acquire knowledge and learn to apply it professionally with due consideration for ethical, ecological and economic issues.
2. **PEO2 (TECHNICAL ACCOMPLISHMENTS):** To provide knowledge based services to satisfy the needs of society and the industry by providing hands on experience in various technologies in core field.
3. **PEO3 (INVENTION, INNOVATION AND CREATIVITY):** To make the students to design, experiment, analyze, and interpret in the core field with the help of other multi disciplinary concepts wherever applicable.
4. **PEO4 (PROFESSIONAL DEVELOPMENT):** To educate the students to disseminate research findings with good soft skills and become a successful entrepreneur.
5. **PEO5 (HUMAN RESOURCE DEVELOPMENT):** To graduate the students in building national capabilities in technology, education and research

PROGRAM SPECIFIC OUTCOMES – Aeronautical Engineering

1. To mould students to become a professional with all necessary skills, personality and sound knowledge in basic and advance technological areas.
2. To promote understanding of concepts and develop ability in design manufacture and maintenance of aircraft, aerospace vehicles and associated equipment and develop application capability of the concepts sciences to engineering design and processes.
3. Understanding the current scenario in the field of aeronautics and acquire ability to apply knowledge of engineering, science and mathematics to design and conduct experiments in the field of Aeronautical Engineering.
4. To develop leadership skills in our students necessary to shape the social, intellectual, business and technical worlds.

SESSION PLANNER

R18A2101- INTRODUCTION TO AEROSPACE ENGINEERING

YEAR: II **SEMESTER: I**

S. NO	UNIT NO	TOPIC	NO OF CLASSES HELD
I	UNIT-1	**History of Flight- the Aerospace Environment**:	
1		Evolution of Flight, Usage of balloons, Dirigibles, Heavier than air aircraft	1
2		commercial air transport, introduction of jet aircraft, helicopters	2
3		Missiles, conquest of space, commercial use of space, exploring solar system and beyond.	1
4		Earth's atmosphere, standard atmosphere	2
5		The temperature extremes of space	2
6		Laws of gravitation, low earth orbit, microgravity, benefits of microgravity	2
7		The near-earth radiative environment. The magnetosphere. Environmental impact on spacecraft	2
8		Meteoroids and micrometeoroids, space debris. Planetary environments.	1 Total 13
	UNIT- II	Aerodynamics and Performance	
1		Airfoil-nomenclature and types;	2
2		Aerodynamic forces on wings and bodies. Generation of lift. Sources of drag.	2
3		Force and moment coefficients, Centre of Pressure. Control surfaces	4
4		Rotary wing aircraft concepts-forces while hovering; Propeller theory	2
5		Performance requirements of civil and military aircraft.	2 Total 12
III	UNIT III	**Propulsion-Aircraft, Rockets and Missiles**	
1		Thrust for flight; Reciprocating engines-two-stroke & four-stroke; the jet engine, rocket engines- description, types, principles of operation.	3

2		Types of missiles; similarities & differences with launch vehicles;	2
3		Control for missiles, airframe components of missiles	2
4		**Evolution of Space missions**: Space missions, mission objectives,	1
		Case studies. Human space flight missions- goals, historical background. The Soviet and US missions. The Mercury, Gemini, Apollo-Soyuz, Space Shuttle	2
		Types of orbits and maneuvers, International Space Station, extravehicular activity. Life support systems.	2
		History and evolution of ISRO; missions carried by ISRO	2 Total 14
IV	**UNIT-IV**	**Structures:** History of airplane construction; Loads on aircraft	**2**
2		Lift production/augmentation devices; Low speed/high speed aerfoils	3
3		Monocoque and semi-monocoque structures, Load bearing structural components	2
4		Use of composites in aircraft and aerospace vehicles	3 Total 10
V	**UNIT- V**		
1		**Experimental Aerodynamics-** Requirement and importance of wind tunnel	2
2		Shock tubes; shock tunnel – types and principle of operation	2
3		Measurement Techniques-Sensors and instrumentation-Pitot static tube	2
4		Cockpit layout of modern civil aircraft	3
5		Basic principles of Gyroscope, accelerometers	4
			Total 13
		TOTAL NO OF CLASSES HELD	**53**

(R18A2104) INTRODUCTION TO AERONAUTICAL ENGINEERING

Objective:

1. Insight overview of various important areas in Aeronautical Engineering
2. Students will acquire the knowledge of the Evolution of Aeronautical industry.
3. To provide an exposure of various forces and performance aspects important for flight.

UNIT - I

History of Flight and Airspace Environment

Evolution of flight, usage of balloons, dirigibles, Heavier than air aircraft, various advances in techniques for commercial transportation. Helicopters, missiles, Conquest of space and exploring solar system and beyond, Earth's atmosphere, standard atmosphere, the temperature extremes of space, laws of gravitation low earth orbit, microgravity, benefits of micro gravity. The near earth Radiative environment, magnetosphere, Environmental impact on spacecraft. Meteoroids, micrometeoroids, space debris and the planetary environments.

UNIT - II

Aerodynamics

Airfoil- nomenclature and types, Aerodynamic forces on wings and bodies, Generation of lift, Sources of drag, moment coefficients; numericals, and centre of pressure.

Rotary wing aircraft concepts – Forces while hovering, Propeller Theory.

UNIT - III

Propulsion- Aircrafts, Rockets and Missiles: Thrust for flight, Reciprocating engines-2 stroke/4 stroke; Jet engine and types, Rocket engines - Description, Principles of operation. Types of Missiles, similarities and differences with launch vehicle, controls for missiles, Airframe components of missiles

UNIT – IV

Aircraft Structures: History of airplane construction, Loads on aircraft, aircraft components, Monocoque and semi-monocoque structures, internal structural members and landing gear, wing and fuselage configurations, engine nacelle, empennage configurations.

UNIT – V

Elements of Aircraft Performance and Measurement Techniques:

Drag Polar, Equations of Motion, Thrust Required for level flight, un-accelerated flight, Thrust available and Maximum Velocity, Power required for

Level and Un-accelerated flight. Power available and maximum Velocity. Altitude effects on Power required and available.

Sensors and instrumentation- Pitot-static tube, Primary Flight Instruments, Basic principles of Gyro, accelerometers.

TEXT BOOKS:

1. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082-4.
2. Kermode, Flight Without Formulae, fifth edition, Pearson Education, 2004, *ISBN*-10: 0273403605; *ISBN*-13: 978-0273403609.
3. V. Ganesan, Gas Turbines, Tata McGraw- Hill, 1999, ISBN, 0070681929

REFERENCES:

1. Turner, M.J.L., Rocket and Spacecraft Propulsion, Springer, 2001.

Outcome:

1. Students acquire knowledge with Aeronautical Engineering to take up study in detail through subsequent courses.
2. Students acquire fundamental concepts of all aspects of flight.
3. Students acquire the knowledge of the important design aspects of aircraft vehicles.

MALLA REDDY COLLEGE OF ENGINEERING AND TECHNOLOGY

(UGC AUTONOUMOUS – Govt. of INDIA)

II B.TECH I SEMESTER – AERONAUTICAL ENGINEERING

Introduction to Aerospace Engineering (R-18)

MODEL PAPER – I

MAXIMUM MARKS: 70

Year: II YR/ Sem I

MODEL PAPER I

ANSWER FIVE QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS

1. Explain the evolution of jet engines and aircrafts

   or

2. Explain the structure of magnetosphere and effects of environmental impact on spacecraft design
3. Explain the various components, types and working principle of a helicopter

   or

4. Explain with a neat sketch the working principle of a jet engine.
5. Consider an aircraft flying approximately at 1500 m altitude with case 1: A_a= 15m^2; c=1.6m; A=2.3m^2; l_t=1600kg.m^2; a_w=5rad^{-1}; $C_{M,ac}$=-0.07; ε=0.45; V=50 m/s; a_t= 4 rad^{-1} ;ρ=1kg/m3 . calculate is the neutral point of a/c.
   case2: let CG is placed halfway between ac and neutral point, x_a=0.21m and $x_{a/c}$=0.132. calculate the angle of attack of the tail and lift produced by tail.

   or

6. Derive the equation of motion for pull up and pull down maneuver.

7. Discuss about the Apollo mission with its phases in mission.

or

8. Explain about power systems and thermal control in a satellite
9. Explain about drawing techniques available

or

10. Discuss about orthographic and perspective projections and discuss their difference.

MODEL PAPER II

ANSWER FIVE QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS 70M

1. Discuss few points on the history of aviation

or

2. Explain about planetary environments within the solar system
3. Derive the equation of buoyancy lift

or

4. Derive the equation of thrust of a jet engine.
5. Define the airplane geometry with a neat sketch and discuss about the two types of stability in detail.

or

6. Derive the equation for maximum lift-to-drag ratio.
7. Discuss about the Gemini mission

or

8. Derive the rocket thrust equation
9. Explain about CAE

or

10. Explain about- Conceptual, preliminary and detail design process.

MODEL PAPER III

ANSWER FIVE QUESTIONS 5X14=70M

1. Explain few points on the advances took during jet engine development

   or

   Explain the types of heat transfer methods and derive expression for equilibrium $P_{emitted}$, $P_{absorbed}$temperature of any body at a distance d from sun.
2. Explain the various types of drag with neat sketches

   or

   Explain with a neat sketch the components & working principle of ramjet engine.
3. Derive the endurance and range equation for a propeller and jet engine.

   or

   derive the equation for trimmed angle of attack.
4. Explain in detail about mercury mission.

   or

   Discuss about the elements of a satellite system.
5. Explain about drawing techniques available

   or

   Discuss about orthographic and perspective projections and discuss their difference.

MODEL PAPER IV

ANSWER FIVE QUESTIONS 5X14=70M

1. Explain the attempts and success during space exploration

   or

   Explain the terms

   a. GCR b. Microgravity c. Solar activity d. Solar flares

2. Explain the terms

   a. Aspect ratio b. Airfoil nomenclature c. Symmetric airfoil d. Cambered

   or

   Derive equation governing propeller propulsion.

3. Derive the equation for longitudinal static stability

   or

   Discuss about accelerated flight and derive required equations.

4. Discuss about the spacesuit design, and discuss the difference between US and Russian.

   or

   with neat sketch explain the working of a ADCS

5. Explain about CAE

   or

   Explain about- Conceptual, preliminary and detail design process.

MODEL PAPER V

ANSWER FIVE QUESTIONS 5X14=70M

1. Explain from the dawn how the improvements that took in aerospace industry from wright flyer to current black bird

   or

   Explain the different layers of atmosphere with a graph showing the temperature and pressure variation

2. Explain the different types of combustor and turbine, derive equations BPR, f, TPR, W_{turb}.

   OR

   a. Explain what is a pitot static tube, position in aircraft and the use.
   b. Using Bernoulli principles derive the equation for stagnation pressure, flow speed.
3. Discuss the V-n diagram with required equations

   or

   Define the airplane geometry with a neat sketch and discuss about the two types of stability in detail.
4. Explain about space shuttle mission and space shuttle with mir, their achievements and mission procedures

   or

   discuss the elements of satellite in detail
5. Explain about LTA design process and point on how LTA takes flight .

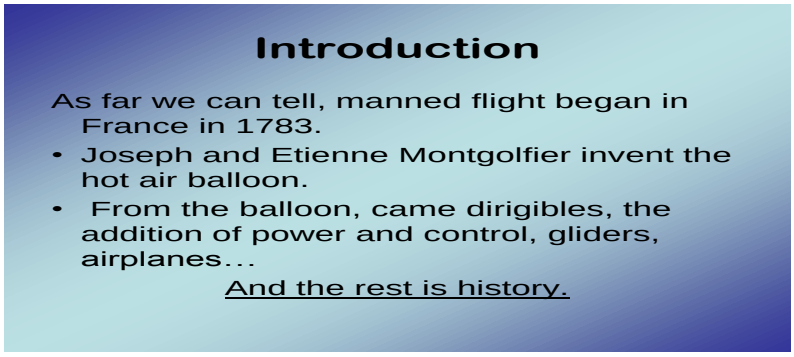
   or

   Explain about ornithopter design and CDR

UNIT - I

History of Flight and Airspace Environment

Evolution of Flight: Usage of balloons, Derigibles, Heavier- than-air aircraft:



Early Aviation Period-1783 – 1914

Lighter-than-air Aircraft:

1. Montgolfier Brothers built the first hot air balloon in April 1783
2. In September 1783 the same balloon lifted a sheep, duck and cockerel into the air
3. By the end of 1783 man had travelled in the balloon, remaining airborne for 25mins
4 J Charles redesigned the balloon and filled it with hydrogen; the balloon's first flight in Dec 1783 travelled 27 miles

Evolution of Flight-Usage ofBalloons, dirigibles-Heavier than air aircraft:

Early Aviation period is from 1783 till 1915.

The development can be grouped as

- Balloons
- Derigibles

- Airships

All man made vehicles which fly and are kept in the air by forces produced by the air, are called aircraft.

Flying Vehicles can be broadly classified as

- Lighter-than-air aircraft – airships, balloons, captive or kite balloons
- Heavier-than-aircraft – power-driven; non power-driven

Manned Flight began in France in 1783. Joseph and Etienne Montgolfier invented the "hot air Balloon"

From the balloon, came dirigibles, the addition of power and controls and other developments.

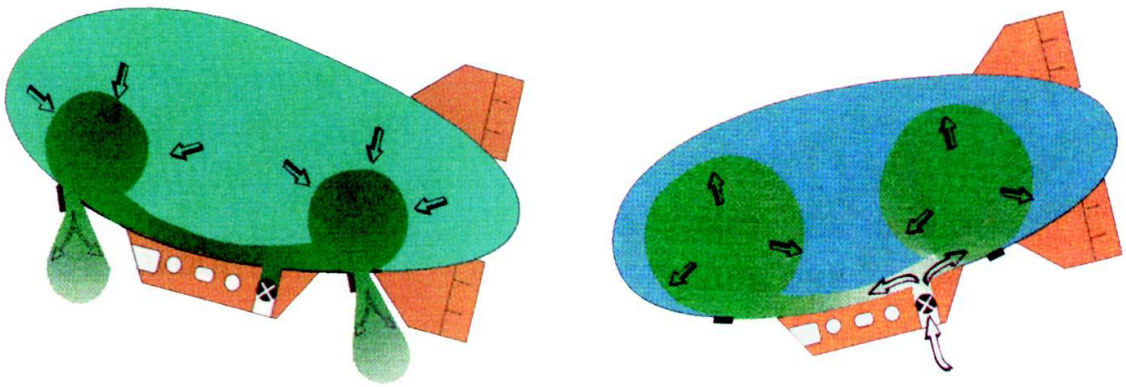
Lighter-than-aircraft: Montgolfier brothers built the first hot air balloon in Apr 1783.

Hot air balloons were very popular till the 1900s; hydrogen was preferred as the filling gas, but, was considered unsafe, causing few explosions.

Balloons (airships) were used to carry cargo, passengers, observation platforms etc.

Controlling airships: Two inflatable air bags were kept inside the main balloon; these inflatable air bags, called "ballonets" are filled with helium. By varying the quantity of helium in the ballonets, the airship is controlled for maneuvering.

By releasing air into the ballonets or by pumping in more air into the ballonets, the helium inside ballonets is compressed or made to expand, thereby causing the airship to raise or descend.

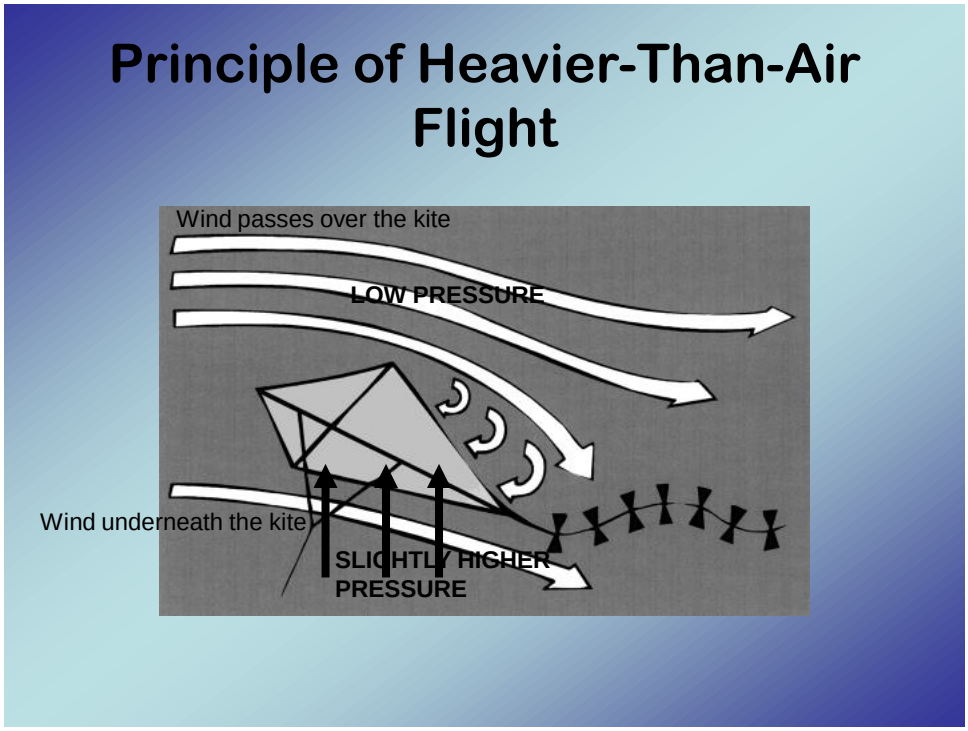


Derigible: An airship or dirigible balloon is a type of aerostat or lighter-than-air aircraft that can navigate through the air under its own power. (shown below)

Zeppelin, rigidairship named after the inventor Zeppelin, consists of a cigar-shaped, trussed, and covered frame supported by internal gas cells.

Heavier-than-air-aircraft- Principle:

- Developed from the flight of a kite
- The shape of kite and its tail enable the kite fly at the correct angle in to the wind
- The weight of the kite is balanced by the force of the wind underneath (Lift)
- The wind passing over the top of the kite creates an area of low pressure
- The air underneath the kite is slightly higher in pressure, so it allows the kite to lift into the lower pressure



- SirGeorgeCayley(1773–1857),devisedthebasicconfiguration ofmodernairplanes.
- Heseparatedthemeansofgeneratingliftfromthemeansof generatingpropulsion.
- In1799,hedesignedanairplanethatfeaturedfixedwings forgeneratinglift,paddlesforpropulsion,andatailunitwithh orizontalandvertical stabilizers.
- In 1804, he built a glider (kite without the string), with no controls
- In 1885, Gottlieb Daimler developed the first single cylinder (combustion engine) aircraft
- In 1903, Wright brothers flew their aircraft

Commercial air transport, introduction of jet aircraft, helicopters:

- Many advancements in aeronautics are aimed at producing powered aircraft that can fly faster and higher.
- The continued push to increase speed has seen the development of the jet engine

- Speeds of conventional aircraft increased from around 100 miles/hour to 2000 miles/hour (F-15) in 1970s.
- Today, the maximum flight speed is pushed to 36,000ft/s (24,500 miles/hour) which is the escape velocity from earth, using Apollo spacecraft
- Most commercial transport aircraft, latest being the Boeing 777 fly at subsonic speeds to minimise the drag.
- Military aircraft are designed not only for high speeds, but high maneuverability (lower speeds increase maneuverability)
- X-15 aircraft became the most notable aircraft flying at over 6.0 M
- The faster & higher concept has been replaced by safer, cheaper, more environmental friendly & long range concept of design keeping the commercial angle of the transport aircraft

Various advances in techniques for commercial air transport: Between World War I till World War II, and during later years, commercial and military aircraft have witnessed many developments to improve speed, height, safety, fuel consumption and endurance/range. Important among these developments are:

1. Multi-cylinder piston-propeller aircraft
2. Turbocharged/supercharged piston engines to improve economical operation and better performance
3. The low speed & low altitude limitation of piston-propeller combination led to discovery of jet engines after 1940s
4. The jet engine which has a compressor/turbine combination has seen development of turbo-propeller, turbojet and turbofan development, which resulted in speeds exceeding speed of sound.
5. The supersonic aircraft were improved with better materials multi-spool compressor-turbine combinations
6. Increased speeds beyond Mach 3.0 resulted in development of Ramjet and later Scramjet aircraft
7. With non-airbreathing engines, the era Rockets began enabling exploration of space

Helicopters:

1. Lift is obtained by means of one or more power driven **horizontal propellers** which called Main Rotor.

2. When the main rotor of helicopter turns, it produces **lift** and **reaction torque**.
3. As the blades of the main rotor spin in the air, an upward lift force is produced.
4. Each blade produces an equal share of the lifting force.
5. Reaction torque tends to make helicopter spin.
6. A small rotor near the tail which called tail rotor compensates for this torque
7. The early helicopters were devised for transportation, but it became quickly apparent that rotary aircraft could also be developed into a formidableweaponssystem.
8. DuringtheVietnam war,BellHelicopterdevelopedthe Bell 209 Huey Cobra attack helicopter, which was the first helicopter designed for such apurpose.

Introduction of Jet Propulsion Vehicles:

Two broad categories:

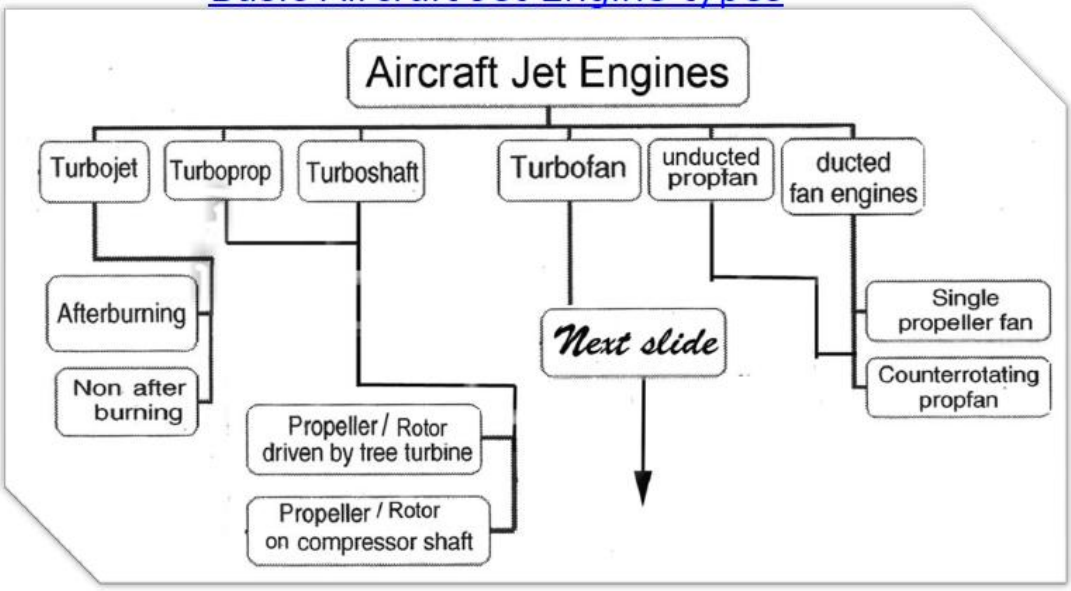
1. Air Breathing Systems: (Reciprocating& Jet Propulsion Engines)

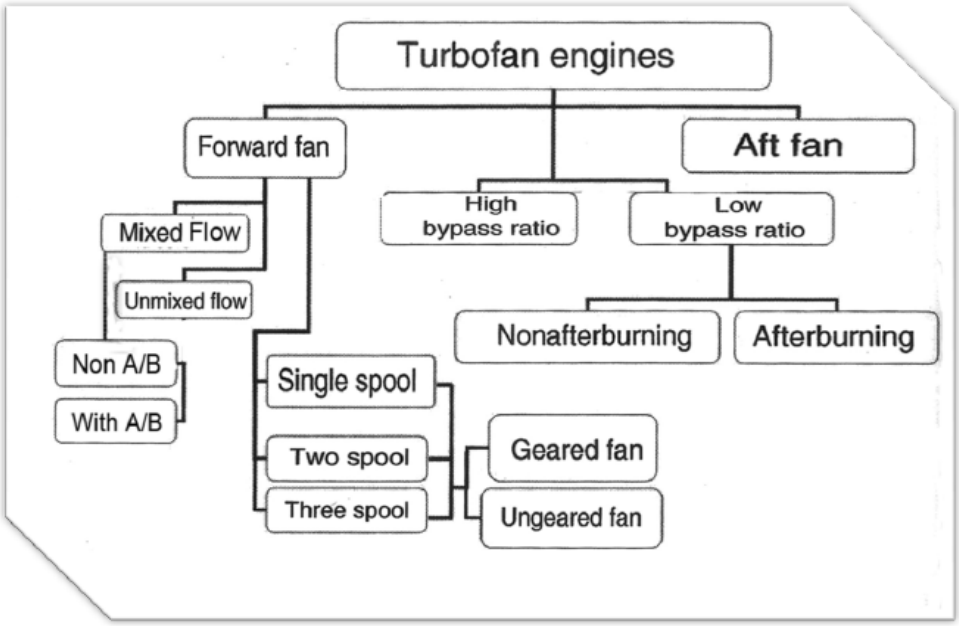
- **Reciprocating Engines**
- **Gas Turbine Engines**
- **Ram Jets, Pulse Jets & Scram Jets**

2. Non Air Breathing Systems

- **Rockets (Chemical, Electric, nuclear &propellant-less)**

Basic Aircraft Jet Engine types





Missiles, conquest of space, commercial use of space, exploring solar system and beyond

In military parlance, missiles are powered / guided munitions are broadly categorised as follows:

- A powered, guided munition that travels through the air or space is known as a military **missile** (or *guided missile*.)

A powered, *un*guided munition is known as a **rocket**

Conventional guided missiles	Cruise missiles	Ballistic misslies
• Air-to-air missile • Air-to-surface missile • Anti-ballistic missile • Anti-satellite weapon • Anti-ship missile • Anti-submarine missile • Anti-tank guided missile • Land-attack missile • Surface-to-air missile • Surface-to-surface missile • Wire-guided missile	• Hypersonic • Supersonic • Long-range subsonic • Medium-range subsonic • Short-range subsonic	• Tactical ballistic missile • Short-range ballistic missile • Theatre ballistic missile • Medium-range ballistic missile • Intermediate-range ballistic missile • Intercontinental ballistic missile • Submarine-launched ballistic missile • Air-launched ballistic missile

The **Solar System** is the gravitationally bound system comprising the Sun and the objects that orbit it, either directly or indirectly. Of the objects that orbit the Sun directly, the largest eight are the planets, with the remainder being smaller objects, small bodies. Moons are the objects that orbit the Sun indirectly.

The **solar system** is made up of the sun and everything that orbits around it, including planets, moons, asteroids, comets and meteoroids

The Solar System formed 4.6 billion years ago from the gravitational collapse of a giant interstellar molecular cloud. The vast majority of the system's mass is in the Sun, with the majority of the remaining mass contained in Jupiter.

The four smaller inner planets, Mercury, Venus, Earth and Mars are terrestrial planets, composed of rock and metal. The four outer planets are giant planets, being substantially more massive than the terrestrials. The two largest, Jupiter and Saturn, are gas giants, being composed mainly of hydrogen and helium; the two outermost planets, Uranus and Nepiune, are ice giants.

All eight planets have almost circular orbits that lie within a nearly flat disc called the ecliptic.

Beyond Solar System:

- The **Universe** is all of space and time and their contents, including planets, stars and galaxies, and all other forms of matter and energy.
- While the size of the entire Universe is still unknown, it is possible to measure the observable universe.
- Our Sun is one of hundreds of billions of stars in a galaxy, named as the "Milky Way". Many of these stars have planets orbiting around them.
- The galaxies are distributed in clusters constituting a vast universe.

Commercial Use of Space:

Commercial use of space is the provision of goods or services of commercial value by using equipment sent into Earth orbit or outer space.

Commercial uses of space include:

1. Satellite Navigation
2. Satellite imagery
3. Satellite television
4. Satellite communication
5. Commercial recovery of space resources
6. Space Tourism
7. Commercial space transportation

Satellite Navigation:

1. A satellite navigation system is a system of satellites in space that provide geo-spatial positioning with global coverage.
2. It allows small electronic receivers to determine their location (longitude, latitude and altitude/elevation) to high precision (within a few metres) using time signals transmitted along a line of sight by radio from satellites.
3. The signals also allow the electronic receivers to calculate the current local time to high precision, which allows time synchronisation.
4. A satellite navigation system with global coverage may be termed a global navigation satellite system (**GNSS**) used in aircraft, ships vehicles and persons in remote areas.

Satellite Imagery:

Earth Observation satellites are specially designed for earth observation from orbit. These are intended for environment monitoring, meteorology, map making etc, Satellite imagery is also used for geological survey, agricultural crop forecasts etc.

Another important military use is spy satellites to detect troop movement, positioning of missiles, monitoring of nuclear weapons etc.

Commercial recovery of space resources:

It includes exploitation of raw materials from asteroids, comets and other space objects, including near earth objects. Minerals and volatiles could be mined then used in space or taken back to earth.

These include gold iridium, platinum, rhodium etc.

Satellite television & communication:

A **communications satellite** relays radio telecommunications signals and creates a communication channel between a source transmitter and a receiver at different locations on Earth. Communications satellites are used for television, telephone, radio, internet and military applications. There are over 2,000 communications satellites in Earth's orbit, used by both private and government organizations.

Space Tourism: For the purpose of pleasure and adventure.

The commercial space transportation industry derives the bulk of its revenue from the launching of satellites into the Earth's orbit. Commercial launch providers typically place private and government satellites into low earth orbit (LEO) and geosynchronous Earth Orbit (GEO). In 2002, commercial space transportation generated 6.6 billion dollars, which made up 6% of the total gross of commercial space activities.

ISRO has been providing satellite launch services to many countries using PSLV & GSLV. It has also developed capability in placing several satellites in different orbits in a single launch.

Communications satellites usually have one of three primary types of orbits

Geostationary satellites have a geostationary orbit, which is 36,000 kilometres from Earth's surface.

This orbit has the special characteristic that the apparent position of the satellite in the sky when viewed by a ground observer does not change, the satellite appears to "stand still" in the sky. This is because the **satellite's orbital period is the same as the rotation rate of the Earth**. The advantage of this orbit is that ground antennas do not have to track the satellite across the sky, they can be fixed to point at the location in the sky the satellite appears.

Medium Earth Orbits (MEO) satellites are around 2,000 to 36,000 kilometres above Earth.The region below medium orbits is referred to as low earth orbits (LEO), about 160 to 2,000 kilometres above Earth.

As satellites in MEO and LEO orbit the Earth faster and do not remain fixed with respect to any one reference point on earth.Therefore, to provide continuous communications capability with these lower orbits requires a larger number of satellites, so one will always be in the sky for transmission of communication signals.

However, due to their relatively small distance to the Earth their signals are stronger

Conquest of Space:

In a world where there are galaxies separated from our own by a distance of six billion light-years, any talk by rocket enthusiasts about "man's conquest of space" seems a trifle silly. Men will land on themoon within the next few years, and within a generation, no doubt, will land on Mars . If there is life on Mars, every round trip by an astronaut will involve grave biological dangers for all concerned. Micro-organisms, to which living things on earth possess no inherited or acquired immunity, may be brought back from our sister planet. Conversely, living things on Mars may succumb to the viruses and bacteria introduced by visitors from Earth. The fruits of this first and, in relation to the whole universe, insignificant "conquest of space" might easily prove to be sudden and irreparable disaster for two biological systems, developed through three or four thousand million years of evolution. And of course the same sort of risks would be run by earthlings visiting any life-supporting globe in any part of the universe.

Conquest of Space:

Six government space agencies re actively involved in space exploration. They are

1. The Indian Space Research Organisation (ISRO),
2. The European Space Agency (ESA),
3. The China National Space Administration (CNSA),
4. The Japan Aerospace Exploration Agency (JAXA),
5. The National Aeronautics and Space Administration (NASA)
6. The Russian Federal Space Agency (RFSA)

Launch Capability: As on date there are nine countries that have orbital launch capability. These are Russia, the United States, France, Japan, China,India, Israel, Iran and North Korea

Mission to Moon: USA sent a manned mission to moon, whereas Russia, China, India and Japan sent unmanned missions to moon.

Satellite Launch: Only **twelve**, countries from the list below (USSR, USA, France, Japan, China, UK, India, Russia, Ukraine, Israel, Iran and North Korea) and one regional organization (the European Space Agency, ESA) have independently launched satellites on their own indigenously developed launch vehicles.

US accounts for approximately one-third of the operational spacecraft currently in orbit around Earth. The Apollo moon-landing missions, the Skylab space station, Space Shuttle, International Space Station (ISS), Mars Exploration Rover – Opportunity, and Mars rover Curiosity are the cornerstones of the nation's space programme.

ISRO has launched over 200 foreign satellites in to orbit.
ISRO's Mars Orbiter Mission (MOM), "Mangalyan" completed 1000 Earth days in Maritian orbit, which went beyond the six-month life spa. MOM entered Maritian orbit on Sep 24, 2014 in its first attempt.
ISRO PSLV-C37 made a world record after launching 104 satellites at once.

Earth's Atmosphere:

The greatest challenges to human spaceflight or high-performance airplane flight are not technological limitations. Rather, the greatest obstacle comes in the human body's ability to successfully tolerate and adapt to the challenges posed by extreme aerospace journeys.

Human evolution in a 1 G (normal Earth gravity) environment under a protective oxygen-rich atmosphere has not prepared the human body for long-duration flight in the hostile environment of space.

Need to study atmosphere:

- Temperature extremes & microgravity effects on equipment and human body
- Radioactive environment in space and its effect on spacecraft and crew
- As space is becoming more crowded, the study of the space environment has evolved to include artificial space debris, along with

(micro) meteorites, and the probability of impacting and damaging an operational space- craft.

Breathable air around earth constitutes mostly oxygen (21%) and nitrogen (78%). Argon (1%), carbon dioxide & water vapor (0.03%). It becomes richer in carbon dioxide and ozone at altitude.

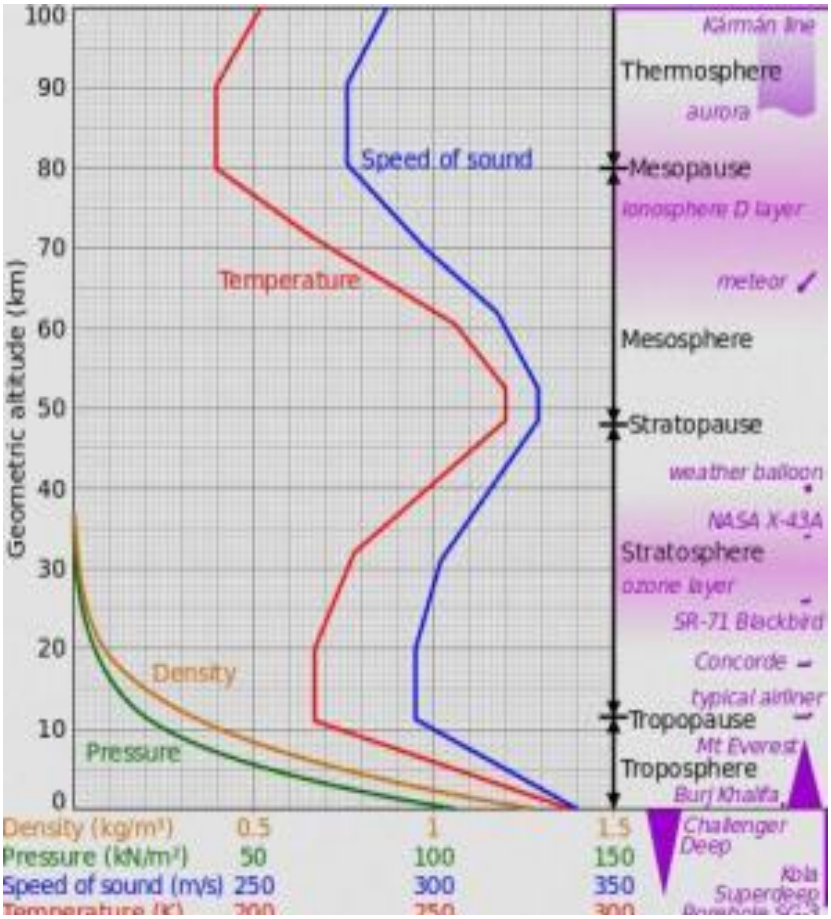
Standard sea-level pressure and temperature are 101.3 kPa and 290 k (17° C) respectively. The density of air on earth 1.22 kg/m^3

The atmosphere is divided in to five parts, namely Troposphere, Stratosphere, Mesosphere, thermosphere and exosphere. Thickness of each part is decided by the temperature profile.

As the altitude increases, the pressure and density decreases so does the thrust. However, as altitude increases, temperature decreases, the thrust increases. The pressure and density decreases faster than the temperature, so the net effect on thrust is to reduce up to an altitude of 11000 (troposphere).

After 11000 m, the temperature stops falling, but pressure continues to drop with altitude. Consequently, above 11000 m, thrust will drop off more rapidly.

This makes 11000 m as optimum altitude for long range cruise.



Troposphere begins at sea level and extends up to 18 km. Temperature which is at 17°C at sea level, drops to - 53°C at 11 km altitude. Between 11 km till 18km, temperature remains constant.

Stratosphere begins at 18 km and extends up to 50 km. The temperature gradient reverses, and the air gets warmer. At 18 km, temperature is around - 53°C and increases to - 3°C at 50 km.

Due to low density, aerospace vehicles can no longer compress air from atmosphere for cabin pressurisation, and there is also a threat from ozone poisoning. Human flights above this altitude must be sealed from environment with independent oxygen and pressure supplies for crew and passengers.

For humans, 22 km might be a good altitude to denote the beginning of space. At approximately 45 km, aircraft propulsion requires independent supply of fuel and oxidizer. So, for aircraft, space begins at 45 km altitude.

Stratosphere also holds the ozone cover, which protects the life on earth from radiation from outer space.

Mesosphere extends from 50 km till 85 km. Temperature decreases to – 83°C at 85 km, making it the coldest region of the atmosphere.
Mesosphere is essential for human survival on earth as it absorbs primary cosmic radiation and the deadly solar ultraviolet and X-ray radiation. It also vaporizes the incoming meteorites entering from interplanetary space.

Beyond the mesosphere the temperature gradient rapidly reverses in the region known as the ***thermosphere***, extending **from 85 to 300 km**. Here the temperature rises from 190 K (—83°C) to 1,000 K (773°C) at higher altitudes where ultraviolet (uv) radiation from the Sun is absorbed.

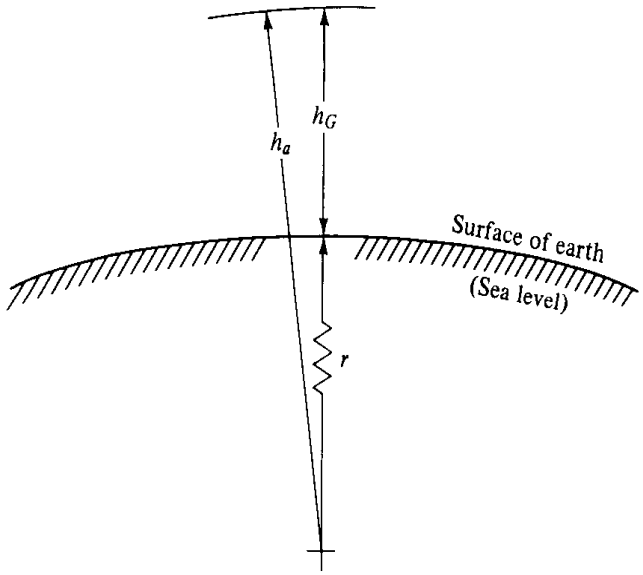
Finally, the ***exosphere*** begins at 300 km and merges with the ionized gases of the inter-planetary medium. The temperature remains constant at1,000 K with the exception of solar cycle variations

Above the exosphere is the region known as *hard space.* At 2,000 km altitude, we still see a density of 10^8 particles per cubic meter. While this number is significantly lower than the density of the surface of Earth, it does indeed show that space is not empty but, rather, is filled with electromagnetic radiation and particles.

Definition of Altitude: The height above ground level is termed as geometric altitude h_G. If radius of earth is r, then the absolute altitude is termed as geometric altitude plus the radius of earth.

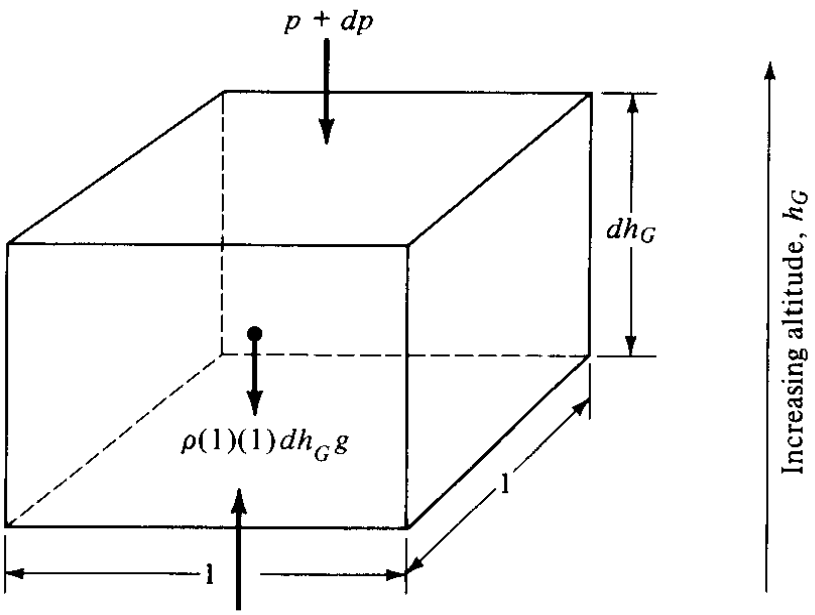
$$h_a = h_G + r.$$

The absolute altitude is important, especially for space flight, since the local acceleration of gravity g varies with h_a.



Hydrostatic Equation:

Force Diagram for Hydrostatic equation:



The hydrostatic equation provides a model that allows calculation of pressure p, density ρ and temperature T as functions of altitude. The equation is nothing but a force balance on an element of fluid at rest.

Consider an element with rectangular faces, where both top and bottom faces have sides of unit length. And are separated by an infinitesimally small height dh_G.

Let the pressure on bottom surface be p, which gives rise to an upward force of p x 1x 1 exerted on the fluid element. The top face is at a slightly higher altitude (by a distance of dh_G). Since the pressure varies with altitude, let the pressure on top face be (p + dp). This pressure gives rise to a force of (p + dp) x 1x 1.

The volume of the element is (dh_G x1x1). Therefore, the mass of the element will be ρ (dh_G x1x1).

If the local acceleration due to gravity is g, then the weight of the finite element will be ρg (dh_G), acting downwards.

The three forces acting on the element will be pressure forces on top and bottom faces and the weight of the element.

The force balance on the fluid element will be

$$P = (p + dp) + \rho g\,(dh_G),$$

thus the change in pressure dp is given by

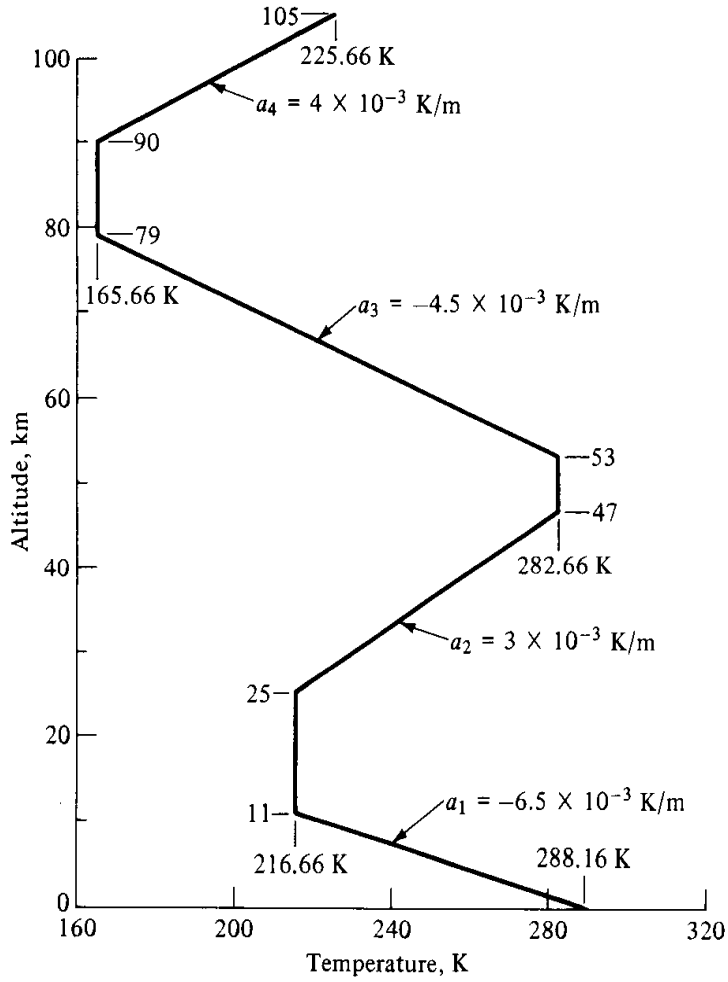
$$dp = -\ \rho g\,(dh_G)$$

This equation is known as the hydrostatic equation.

The acceleration due to gravity varies very slightly up to a height of 15 km above sea level. Therefore, the hydrostatic equation may be written as

$dp = -\ \rho g_o\,(dh_G)$, where g_o is the acceleration due to gravity at sea level.

Temperature distribution with altitude:



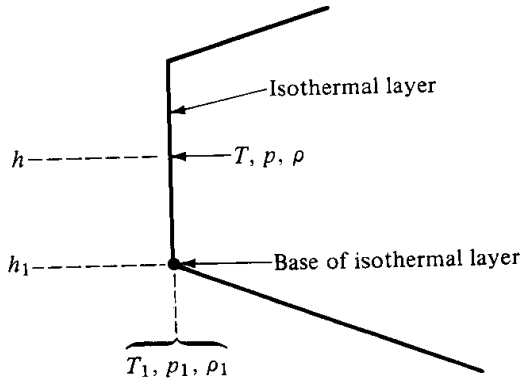
Relation between p, T and ρ as functions of altitude: The keystone of the standard atmosphere is a defined variation of T with altitude.

The variation of T with altitude consist of a series of straight lines. Some are vertical (called isothermal regions where T remains constant) and some are inclined lines, called gradients, where T varies linearly with some lapse rate.

We are interested in altitudes up to 25 km, where T reduces linearly between 0 – 11 km, and remains constant between 11 - 25 km.

Consider the isothermal region: (11 – 25 km)

T is 216.66 k at 11 km and remains same up to 25 km.



As per hydrostatic equation, dp = − ρg_0 dh

Dividing with equation of state, p = ρRT, we get

$$\frac{dp}{p} = -\frac{\rho g_0 dh}{\rho RT} = -\frac{g_0}{RT}dh$$

Integrating the above equation, we get relation between ρ_1, p_1 and T_1 at the base of the isothermal (ie at 11 km) and ρ, p and T, at height between 11 – 25 km altitude.

$$\int_{p_1}^{p}\frac{dp}{p} = \frac{g_0}{RT}\int_{h_1}^{h} dh$$

Integrating, we get,

$$\boxed{\frac{p}{p_1} = e^{-(g_0/RT)(h-h_1)}}$$

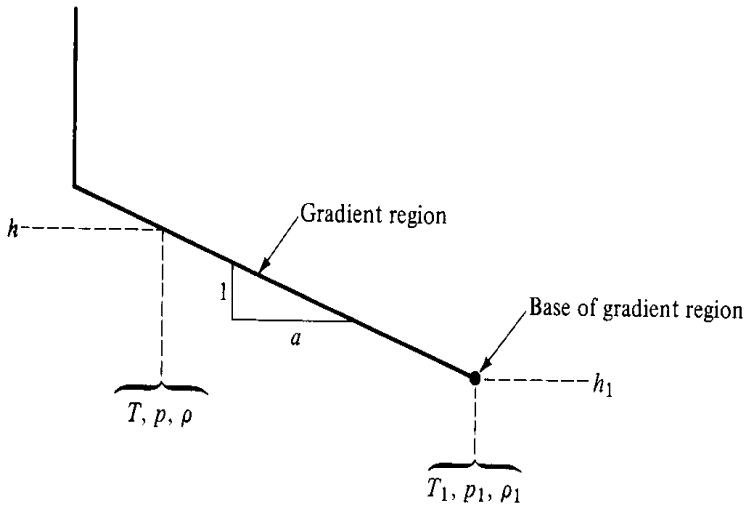
And for density,

$$\frac{p}{p_1} = \frac{\rho T}{\rho_1 T_1} = \frac{\rho}{\rho_1}$$

$$\boxed{\frac{\rho}{\rho_1} = e^{-(g_0/RT)(h-h_1)}}$$

Above equations give variation of density and pressure in isothermal regions. We use these equations to find ρ and p between 11 – 25 km.

Consider gradient region 0 – 11 km altitude.



The temperature variant with height

$\frac{T - T_1}{h - h_1} = \frac{dT}{dh} = a$

$$dh = \frac{1}{a} dT$$

we know that

$$\frac{dp}{p} = -\frac{g_0}{RT} dh$$

Substituting for the value of dT, we get

$$\frac{dp}{p} = -\frac{g_0}{aR}\frac{dT}{T}$$

Integrating between heights $h_1$and any height h, between 0 – 11 km altitude, we get

$$\boxed{\frac{p}{p_1} = \left(\frac{T}{T_1}\right)^{-g_0/aR}}$$

And from equation of state, $p/p_1 = \rho T/\rho_1 T_1$, we get,

$$\boxed{\frac{\rho}{\rho_1} = \left(\frac{T}{T_1}\right)^{-[(g_0/aR)+1]}}$$

Between heights 0 – 11 km, the variation of T is linear and is given by

$$\boxed{T = T_1 + a(h - h_1)}$$

The base values of pressure, density and temperature at sea level condition is

$$p_s = 1.01325 \times 10^5 \text{ N/m}^2 = 2116.2 \text{ lb/ft}^2$$

$$\rho_s = 1.2250 \text{ kg/m}^3 = 0.002377 \text{ slug/ft}^3$$

$$T_s = 288.16 \text{ K} = 518.69°\text{R}$$

mperature at 11 km – 216.66 K

Temperature lapse rate "a" between 0 – 11 km = – 0.0065 K/m

Problem: Calculate the standard atmosphere values of T, p and ρ at an altitude of 14 km.

Solution: T_s= 288.16 K

Lapse rate "a' = 0.0065 K/m

Temperature at 11 km = 288.16 – (0.0065 X 11000) = 216.66 K

We know that at 0 km altitude, p_1= 1.01 X 10^5N/m^2 and ρ_1= 1.23 kg/m^3,

At 11 km, we now find the pressure and density;

P_{11}= p $(T/T1)^{-g0/aR}$ = (1.01 X 10^5) $(216.66/288.16)^{(-9.8/-0.0065 \text{ X } 287)}$

Which gives p at 11 km = 2.26 X 10^4 N/m^2

Density at 11 km, ρ_{11}= ρ_1 $(T/T_1)^{-(g_0/aR)+1}$

Which gives ρ_{11}= 0.367kg/m^3

The above values of p and ρ for the isothermal region between 11 – 25 km.

To find pressure and density at 14 km:

T_{14}= 216.66 K

p_{11}, and $\rho_{11}$are known. Find pressure and density at 14 km using following equations:

$$\boxed{\frac{p}{p_1} = e^{-(g_0/RT)(h-h_1)}}$$

$$\frac{p}{p_1} = \frac{\rho T}{\rho_1 T_1} = \frac{\rho}{\rho_1}$$

$$\boxed{\frac{\rho}{\rho_1} = e^{-(g_0/RT)(h-h_1)}}$$

Temperature extremes of Space:

The extreme thermal conditions in space require not only shielding and insulation, but heat rejection capabilities as well.

In the vacuum of space, an isolated body, such as a space-suited astronaut during a spacewalk or a planet or a satellite, can transfer heat to or from another body solely by radiation, provided the body is in a state of thermal equilibrium.

The physical phenomenon of radiation is governed by Kirchoff's law, which states that a body in thermal equilibrium will radiate an amount of energy equal to that absorbed from the outer universe.

Energy exchange and balance determines the temperature of an astronaut, planet or satellite.

The extent to which a body absorbs solar radiation is determined by the *solar absorption coefficient a*.

The power absorbed by a body in space, from Sun, is directly proportional to the solar absorption coefficient a, square of the radius of the body and inversely proportional to the distance d of the body from Sun.

The power emitted by the body, is linked to its equilibrium temperature T through s, the Stefan–Boltzmann constant.

Equilibrium temperatures of various bodies in solar system are thus calculated. The equilibrium temperature T of Earth is 257 k; Mars is 217 k and Jupiter is 107 K.

Microgravity:

The term "microgravity" is used to describe a very low-acceleration environment. In a low earth orbit, the spacecraft and astronauts experience a radial gravity effect that is only one-tenth of the standard 9.8 m/s^2, (1g)

environment. While orbiting earth, the spacecraft experiences a constant state of "free fall".

The term **micro-g environment** is more or less a synonym of *weightlessness* and *zero-G*, but indicates that g-forces are not quite zero, just very small.

Benefits of Microgravity:

1. Microgravity allows scientists to observe phenomenon usually overshadowed by gravity on surface of earth.
2. Medical professionals today use medicines developed in space environment.
3. Material scientists and engineers have better understanding of how substance interact to form various materials (alloys)
4. Working in microgravity environment allows researchers to investigate essential questions of fundamental physics, life sciences, materials, medicine and biological sciences.
5. Engineers explore new technologies and develop devices specially designed to function in microgravity environment.

Low Earth Orbit:

Low Earth orbit, or LEO, is the term used to define orbits used for human spaceflight and some satellites, which are rather close to Earth (~300 to 1,000 km altitude). By flying in an orbit close to Earth, astronauts avoid strong doses of radiation that result from flying in high-altitude orbits. LEO space travel does place some limitations on the mission and the functions of the satellite in orbit.

Satellites orbiting around Earth in LEO are balanced by the outward-pulling centrifugal force and inward-pull of Earth's gravitational attraction. Even though LEO is considered as low acceleration zone, many forces like vibrations due to orbital thrusters, aerodynamic drag etc act on the spacecraft. Earth orbiting satellites follow elliptical paths.

Heavy radiation is major concern for human spacecraft missions.

Space shuttle and other vehicles use LEO to slow down their speed due to the aerodynamic drag.

Laws of Gravitation:

Kepler defined the laws of orbital motion while Sir Isaac Newton defined the Law of Gravitation.

As per law of universal gravitation, every particle in universe attracts every other particle with the force that is directly proportional to the product of their masses and inversely proportional to the distance between their centres.

$$F = G \times \frac{m_1 \times m_2}{r^2}$$

Where F is the force due to mass m_1 on mass m_2, and r is the vector from m_1 to m_2; G is the universal gravitational constant whose value is 6.67 X 10^{-8} dyn/cm^2/g^2).

The Near Earth Radiative Environment:

The Planet Earth offers its inhabitants a shield from hostile space environment, which consists of the atmosphere and its magnetic field.Outside this double protection, various types of radiations are encountered. They are briefly discussed below:

Solar activity & radiations: Sun is a modest star compared to more than 100 billion stars that form our galaxy. However, Sun provides all heat input to the solar system and dominates the gravitational field. The Sun has no distinct surface or discrete physical boundary.

- The gravity of Sun creates extreme pressures and temperatures within itself, which makes it fundamentally a giant thermonuclear fusion reactor. This causes regions of extremely strong magnetic fields around Sun. Dense clouds of material is suspended above the surface of the Sun by loops of the magnetic field. This is observable as a halo around the Sun and termed as “Corona”
- **Solar Wind-** The Corona forms an external gaseous envelope around Sun, with extremely high temperatures. The Corona continuously ejects particles, mainly electrons and protons. This continuous flux of charged particles constitutes “solar wind”, which streams off in all directions at **400 km/sec** speeds.The temperature of Corona is so high, that Sun’s gravity cannot hold it. The charged particles diffuse and spread in the entire inter-planetary space.
- **Sunspots & Sun Flares**: Sunspots are places where very intense magnetic lines of force break through the Sun’s surface. Sun Flares arise

out of sun spots, causing violent release of energy for short durations. The burst of energy produce various radiations, mainly X-rays and gamma rays into the interplanetary regions.

- **Gelactic Cosmic Rays (GCR)**: GCR rays are mainly protons, heavy ions and alpha particles with extremely high energies. These are emitted by distant stars and even more distant galaxies. GCRs diffuse through space and arrive at Earth in all directions. GCRs are dangerous to spacecraft and humans inside, because a single particle can damage common electronic equipment on-board. LEO orbits experience less threat from GCRs due to strong shielding provided by the Earth's atmosphere and geomagnetic field.

- **Van Allen Belts**: These are doughnut-shaped regions of trapped particles of protons and electrons, within LEO orbi (at around 1400 km) which provide severe environment for space craft and human missions passing through them.

Magnetosphere: The magnetosphere is defined by the interaction of Earth's magnetic field and solar wind. The effect of radiations/flares/solar wind from Sun are felt in magnetosphere. The depth of magnetosphere is up to 10 "Earth radii", around Earth, covering LEO, GEO and Geo-stationary orbits. The field lines of magnetosphere originate from North pole to South pole.

The radiation particles interfere with satellite, aircraft and spacecraft communications in LEO causing potential damage to spacecraft communications causing blackouts (no radio signals being received by satellites etc). These blackouts may last about 15 to 30 min in each orbit while the satellite is passing through areas of high particle concentration. Even the communication satellites orbiting in Geostationary Earth orbit (GEO) may be effected by the particle interference.

Additionally, humans in space also are likely to be effected by the high doses of radiation.

Radiation can severely degrade optical, mechanical and electrical properties of a spacecraft. Environmental parameters considered while designing spacecraft include "absorbed dose of radiation", electrostatic arcing due to accumulation of charges and transient effects.

The absorbed dose is defined as the ratio of the average energy transferred to a given volume of a material by the radiation, per mass of this volume

The absorbed dose of radiation impacts the on-board electronics resulting in loss of functionality.

Thecumulativedoseeffectismainlyconsideredinspacecraftdesignbecause ofitsimpactontheon-boardelectronics.Semi-conductorcomponentsandbipolar structures undergo a severe degradation of their key parameters with increasing dose, resulting in functional failures

Single Energy Particle Effects (SEE) are radiation events caused by a single energetic particle (galactic cosmic rays, solar protons, trapped particles in the Van Allen belts), which are most damaging to electronic components.

Electrostatic charging of a spacecraft on orbit always occurs. Some materials behave differently on the day side of the orbit than during eclipse, they accumulate charges during eclipse which alters their operational characteristics.

Environmental impact on spacecraft:

Space environment is a branch of astronautics, aerospace engineering and astronomy that seeks to understand and address conditions existing in space that would impact both the operation of spacecraft and also affect our planet's atmosphere and geomagnetic field.

Problems for spacecraft can include radiation, space debris, upper atmospheric drag, and the solar wind. Effects on Earth of space environmental conditions can include ionospheric storms, temporary decreases in ozone densities, disruption to radio communication, to GPS signals and submarine positioning. Some scientists also theorize links between sunspot activity and ice ages.

Solutions explored by scientists and engineers in the area of space environment study include, but are not limited to, spacecraft shielding, various collision detection systems, and atmospheric models to predict drag effects encountered in lower orbits and during re-entry.

The field often overlaps with the disciplines of astrophysics, atmospheric science, space physics, and geophysics, albeit with a stronger emphasis on application.

The United States government maintains a Space Environment Centre at Boulder, Colorado. The Space Environment Centre (SEC) is part of the National Oceanic and Atmospheric Administration (NOAA). SEC is one of the

National Weather Service's (NWS) National Centres for Environmental Prediction (NCEP).

Meteoroids, micrometeoroids and space debris. Planetary environments:

Meteoroids:

1. Interplanetary meteoroids are solid objects of sizes 10^{-15} kg (Very small) to 10^{13} kg (very large).
2. The primary concern with meteoroids is their probability of collision with the spacecraft, which can be fatal.
3. If a meteoroid reaches the ground surviving the impact, then it is called a meteorite.

Micrometeoroids:

1. Micrometeoroidsare usually solid particles with a mass range from 10^{-15} to 10^{-1}kg.
2. A micro-meteoroid has sufficient energy to penetrate the protective coatings on satellites.
3. Micrometeoroids pose significant threat for space exploration
4. Their velocities in space are in the order of km/sec can make significant damage to spacecraft on impact
5. Spacecraft and space suit designers need to take this aspect while designing the equipment.

Effect of environment on Spacecraft:

- Micrometeoroids pose a significant threat to space exploration.
- Their velocities relative to a spacecraft in orbit can be on the order of km per second, and resistance to micrometeoroid impact is a significant design challenge for spacecraft and space suit designers
- While the tiny sizes of most micrometeoroids limits the damage incurred, the high velocity impacts will constantly degrade the outer casing of spacecraft in a manner analogous to sandblasting.
- Long term exposure can threaten the functionality of spacecraft systems.

Space Debris:

Orbital debris generally refers to material that is on orbit as the result of space initiatives,butnolongerservesanyfunction.Therearemanysourcesofdebris. One source is discarded hardware. For example, many upper stages from launch vehicles have been left on orbit after they are spent. Many satellites are also abandoned after end of usefullife.

Another source of debris is spacecraft and mission operations, such as deploymentsandseparations.Thesehavetypicallyinvolvedthereleaseofitems suchasseparationbolts,lenscaps,momentumflywheels,nuclearreactorcores, clamp bands, auxiliary motors, launch vehicle fairings, and adaptershrouds.

A major contributor to the orbital debris background has been object breakup. More than 124 breakups have been verified, and more are believed to have occurred. Breakups generally are caused by explosions and collisions.

The majority of breakups have been due to explosions. Explosions can occur when propellant and oxidizer inadvertently mix, residual propellant becomesover-pressurizedduetoheating,orbatteriesbecomeover-pressurized. Some satellites have been deliberately detonated. Explosions can also be indirectly triggered by collisions withdebris.

Orbitaldebrisgenerallymovesatveryhighspeedsrelativetooperationalsatellites.InlowEarthorbit,i.e.,ataltitudeslessthan2,000km,theaveragerelative velocity at impact is 10 km per second.Atthisvelocity,evensmallparticlescontainsignificantamountsof kinetic energy and momentum.

Debris fragments from 1-10 cm size will penetrate and damage most spacecraft. If the spacecraft bus-bar is impacted, the satellite function will be terminated. At the same time, a significant amount of small debris will be created. In large satellite constellations, this can lead to amplification of the local smaller debris and associated erosional effect.

If a 10 cm debris fragment weighing 1 kg collides with a typical 1200 kg spacecraft, over one million fragments of 1mm size and larger can be created. This results in the formation of a debris cloud which poses a magnified impact risk to any other spacecraft in the vicinity.

Debris Mitigation Efforts: Risk is best prevented controlled by limiting the creation of debris through mitigation. Unfortunately, debris mitigation usually increases mission costs.

Some debris mitigation efforts involve **simple changes in design**, like tethered lens caps and bolt catchers which prevent ejection of objects. The propellants left unburnt in upper stages and satellites after useful life can be eliminated by venting or burning. Batteries can be designed to reduce risk of explosion.

Satellites and other objects must be removed from mission orbit after completing useful life. NASA stipulates maximum time span of 25 years for orbiting objects.

Debris in LEO at sufficiently low altitudes, atmospheric drag on the objects causes decay of the orbit gradually, and result in burning the objects on re-entry.

Vehicles at higher altitudes can perform post-mission maneuvers to drop to lower orbits. Propellant must be catered for these maneuvers at the end of operational life.

At altitudes above 2000 km, it is not feasible to bring down the debris objects. Such objects are placed in disposal orbits (often called graveyard orbits).

Planatory environment:

All the larger planets have atmospheres. The atmospheres of planets are held in place only by the force of gravity and do not have distinct boundaries in space. Therefore, a spacecraft operating in the vicinity of a planet with an atmosphere mustbedesignedtotoleratetheeffectsofthatatmosphere,particularlyifhumans are aboard thevehicle.

The chemical composition of a planetary atmosphere varies greatly among planets depending on the distance from the Sun, its size, and its unique large-scale chemical reactions. The inner Earth-like planets (Mercury to Mars) are believed to have formed without extensive atmospheres. One theory as to how atmospherescameaboutisthatgasesoncechemicallycombinedinthecrusthave since escaped, and that additional gases from comet impacts became trapped under the gravity of the planet.

In the case of Earth, research has shown that the presence of uncombined oxygen in noticeably large quantities is a direct result of the presence of life and suggests that the current atmosphere has evolved significantly over time (the past 4 billion years). Conversely, studies of the outer planets (Jupiter to Pluto)

suggest that the atmospheres present today are nearly identical to those present when the planetsformed.

UNIT - II

Aerodynamics

Airfoil- nomenclature and types, Aerodynamic forces on wings and bodies, Generation of lift, Sources of drag, moment coefficients; numericals, and centre of pressure.

Rotary wing aircraft concepts – Forces while hovering, Propeller Theory.

Aerodynamics

Airfoil- nomenclature and types, Aerodynamic forces on wings and bodies, Generation of lift, Sources of drag, moment coefficients; numericals, and centre of pressure.

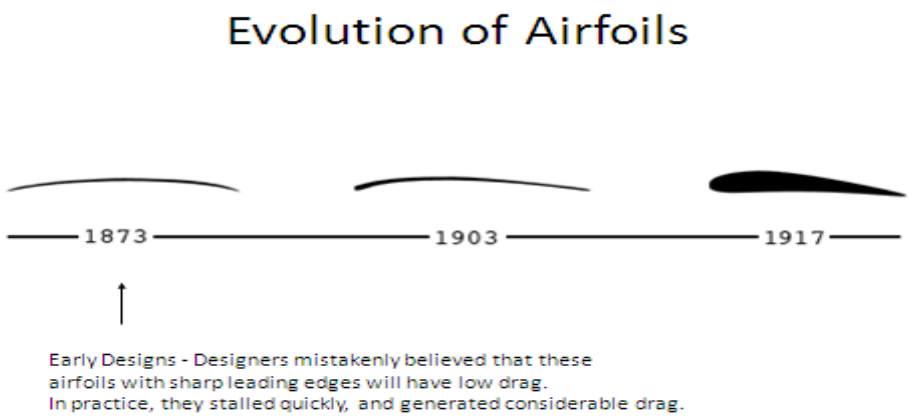
Rotary wing aircraft concepts – Forces while hovering, Propeller Theory.

Airfoil – Nomenclature and types:

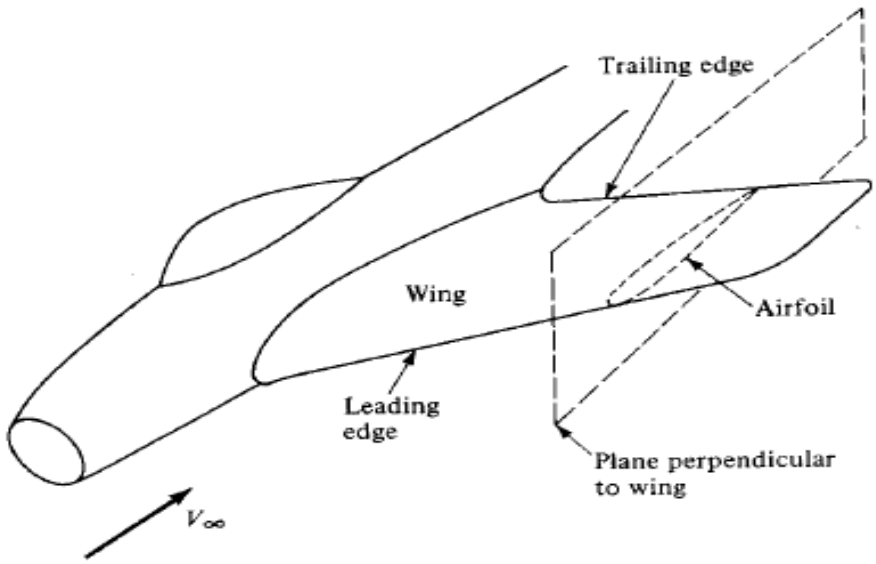
Uses of airfoil sections:

- Wings
- Propellers and Turbofans
- Helicopter Rotors
- Compressors and Turbines
- Hydrofoils (wing-like devices which can lift up a boat above waterline)
- Wind Turbines

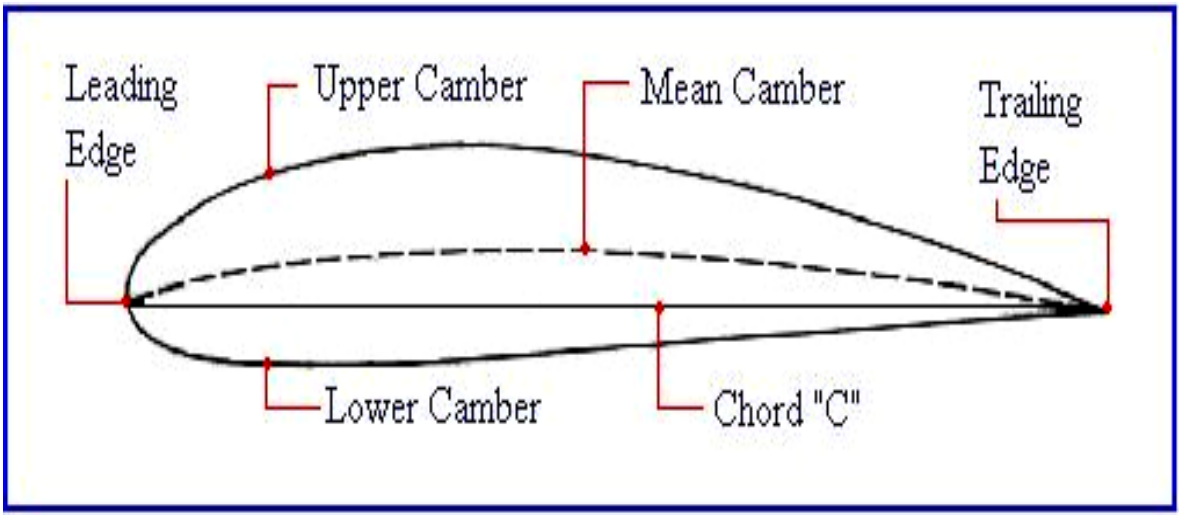
The Bernoulli equation describes lift as generated by a pressure difference across the wing, and Newton's law states that lift is the reaction force on a body caused by deflecting a flow of gas. A body moving through a fluid generates a force by the pressure variation around.



Airfoil:



Airfoil is a section of the wing of an airplane that is designed to produce lift.



An airfoil is defined by:

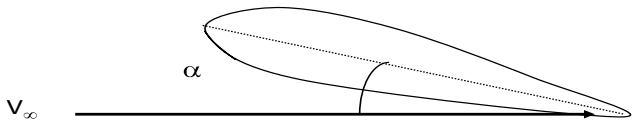
- Chord Line
- Camber line drawn with respect to the chord line.
- Thickness Distribution which is added to the camber line, normal to the camber line.
- Symmetric airfoils have no camber.

Airfoil-Definitions & nomenclature

- The straight line connecting the leading and trailing edges is the **chord line**.
- The distance from the leading to trailing edge measured along the chord line is called the **chord** of the airfoil, denoted as *c*.
- **Mean Camber Line** is the locus points halfway between the upper and lower surfaces as measured perpendicular to the chord line.
- **Angle of attack** is defined as the angle between the freestream and the chord line. It is given the symbol α

The freestream velocity V_∞ is the velocity of the air far upstream of the airfoil. The direction of V_∞ is defined as the relative wind.

Angle of Attack



•**Angle of attack** is defined as the angle between the freestream and the chord line. It is given the symbol α.
• The angle of attack may change from root to tip.
•The root will, in general, have a high angle of attack.
•The tip will, in general, have a low (or even a negative) α.

Types of Airfoils

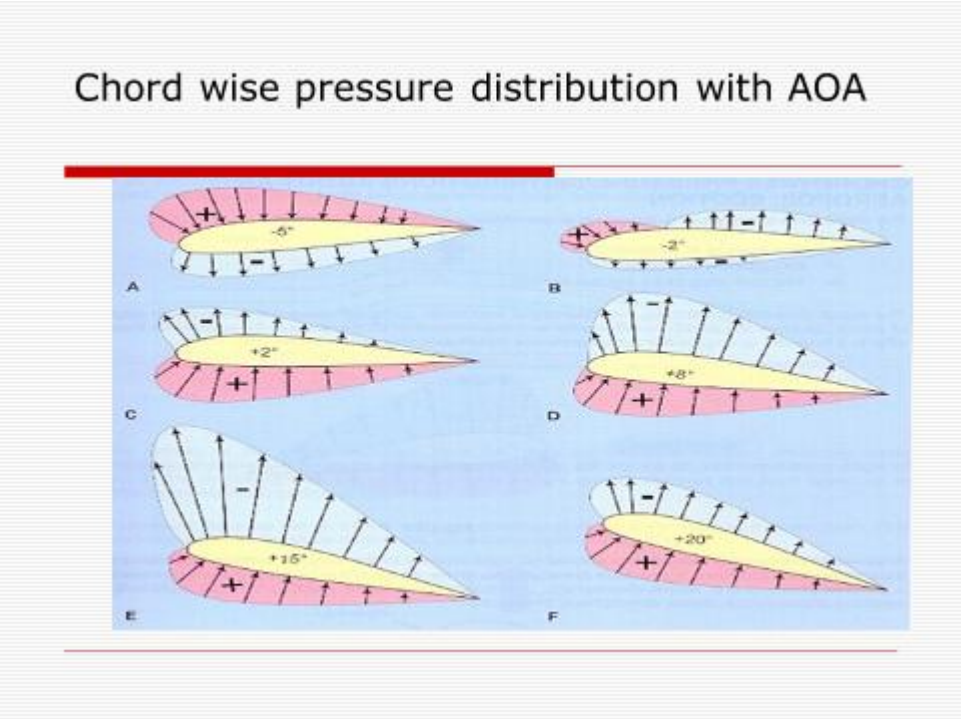
Airfoil Development: Between 1910-1920, British (RAF)and USA (NPL) led in developing airfoils of different cambers. Later, National Advisory Committee for Aeronautics (NACA) developed "NACA four-digit airfoils".

NACA four-digit airfoils: The four digits of the airfoil indicate the extent and location of maximum camber and maximum thickness. For example the NACA 4412 airfoil is defined as a shape that has

- Maximum camber of 2 percent of the chord (first digit)
- Location of maximum camber occurs at 0.4 percent chord from the leading edge (the second digit)
- The maximum thickness of 12 percent of chord (last two digits)

NACA five-digit airfoils: NACA developed new camber family line to increase lift considerably, which is the 230 series camber. The new five-digit airfoils belong to 230 family (the first three digits of 230) followed by the magnitude of maximum thickness. For example, the NACA 23012 airfoil belongs to 230 camber family with 12 percent maximum thickness.

- Thus NACA 2412 has 2% camber at 40% chord from LE and is 125 thick (max)
- NACA 23012 has C_l of 0.3 (2 X 0.15) with max camber at 15% chord from LE and is 12% thick (max),
- NACA 632-315 is 6-series with minimum pressure 30% of chord back from LE, design C_l of 0.3 ± 0.2 and is 15% thick (max)



With the advent of space programs, USA has set up National Aeronautics and Space administration (NASA) in 1958 and wound up NACA.

Since 1958, NASA has been in charge of developing new technologies for aircraft and also space vehicles.

Theoritical aerodynamics finds the values of aerodynamic forces and coefficients using mathematical equations, using some simplifying assumptions. The values found through these mathematical models are often not accurate.

Practical experimental methods are widely used to measure forces and coefficients for specific bodies of interest.

NACA has compiled a large bulk of experimental data on airfoils of different shapes. Later, the work continued by NASA. Lift, drag and moment coefficients were measured on many airfoil shapes under different flow conditions at subsonic, supersonic and higher speeds.

Aerodynamic Forces and Moments:

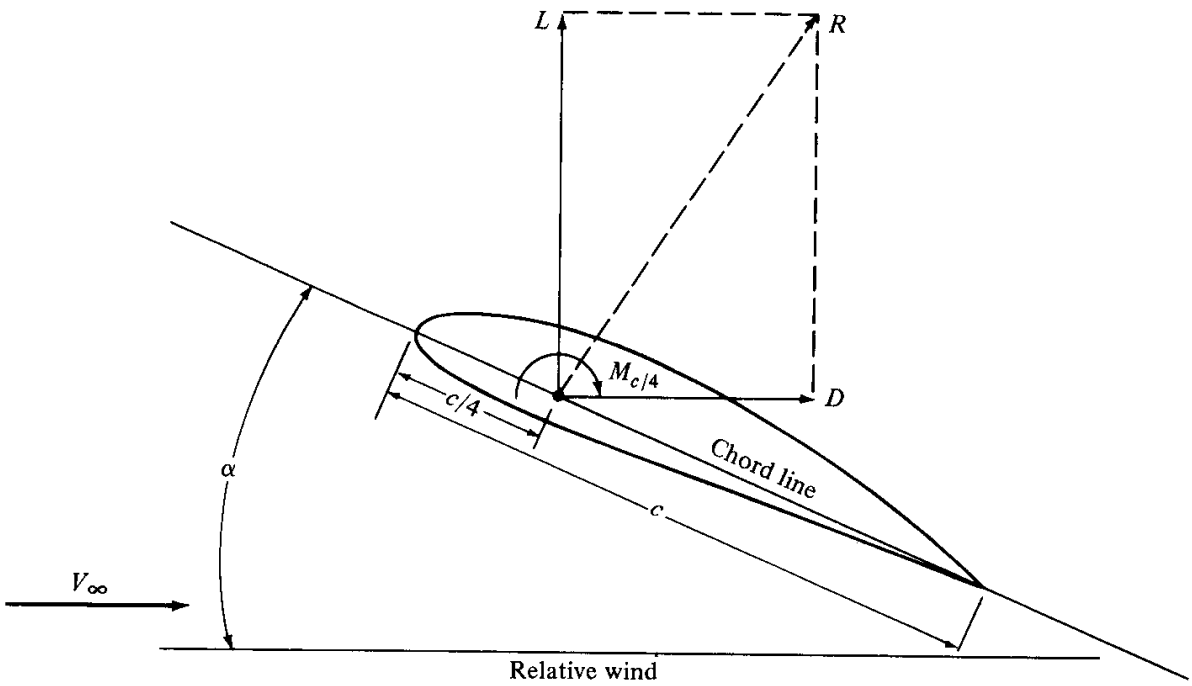
The aerodynamic forces and moments on a body immersed in a fluid are due to two basic sources:

- Pressure distribution over the body surface
- Shear stress distribution over the body surface

The pressure distribution p acts normal to the surface of the body, and the shear stress τ acts tangential to the surface of the body.

The net effect of the p and τ over the complete body is to generate a resultant aerodynamic force R and moment M on the body. The pressure differential over the airfoil gibes raise to the resultant force R as shown below:

Sketch showing Lift, Drag, Moment, angle of attack and relative wind direction:



The wings provide lift by creating a situation where the pressure above the wing is lower than the pressure below the wing. Since the pressure below the wing is higher than the pressure above the wing, there is a net force upwards.

The lift L is always defined as the component of aerodynamic force perpendicular to the relative wind direction.

The drag D is always defined as the component of aerodynamic force parallel to the relative wind direction.

The shear force distribution over the airfoil create a moment M which tends to rotate the wing. It is most common in case of subsonic airfoils, to take the moment about a point on the chord at a distance c/4 from the leading edge, the "quarter-chord point". This moment is designated as $\mathbf{M_{c/4}}$.

There exists a point on airfoil along the chord line, about which the moment will not vary with angle of attack. This point is designated as aerodynamic center and the moment about the aerodynamic center is designated as $\mathbf{M_{ac}}$.

Lift, Drag and Moment Coefficients:

The sources of aerodynamic forces lift, drag and moments on the body are the pressure and shear stress distributions integrated over the body. The aerodynamic forces and moments depend on quantities that are very

fundamental in nature. These are the dimensionless force and moment coefficients.

The lift, drag and moment coefficients are defined by a basic quantity called dynamic pressure q_∞, which depend on free stream conditions and flight speed.

$$q_\infty = \frac{1}{2}\rho_\infty V_\infty^2$$

The lift coefficient c_l is defined as aerodynamic force divided by the dynamic pressure and some reference area (for wings, convenient reference area is S)

$$C_l = \frac{L}{q_\infty S}$$

Drag Coefficient is defined as $C_d = \frac{D}{q_\infty S}$, and

Moment coefficient $C_m = \frac{M}{q_\infty S l}$

The symbols in capital letters indicate force and moment coefficients for the entire three-dimensional body. For two-dimensional bodies, it is conventional aerodynamic coefficients in lowercase letters, as c_l, c_d and c_m.

Defining values of lift, drag and moment coefficients: The lift, drag and moment on an airfoil vary based on the following parameters;

- Freestream velocity V_∞
- Freestream density ρ_∞, at given altitude
- Size of aerodynamic surface. For airfoils, area of wing "S" is used to indicate surface area
- Angle of attack, α
- Shape of airfoil, indicated by the camber
- Viscosity coefficient μ_∞, since aerodynamic forces are generated in part by the skin friction distributions
- Compressibility of the flow. Governed by speed of sound a_∞.

The lift force $L = f(V_\infty, \rho_\infty, S, \mu_\infty, a_\infty)$,

Through a dimensional analysis, we can representLas

$L = Z\, V^{a} \rho^{b} S^{d} a^{e} \mu^{f}$, where Z, a, b, d, e, f are dimensionless constants. We also define a non-dimensional number called "Reynolds number" to represent the viscous forces.

Using dimensional analysis, equating the physical quantities on both sides with respective dimensions, we get,

$$\frac{ml}{t^2} = \left(\frac{l}{t}\right)^a \left(\frac{m}{l^3}\right)^b (l^2)^d \left(\frac{l}{t}\right)^e \left(\frac{m}{lt}\right)^f$$

The exponent of m on left hand side is 1; and on the right hand side is b+f. We can write

1 = b + f, similarly, we will equate exponents of time t and length l, we get following equations;

□ 2 = □ a □ e □ f

1 = a □ 3b + 2d + e □ f

solving the above equations, we get

b = 1 □ f

a = 2 □ e □ f

d = 1 □ $\frac{f}{2}$

substituting these values in the equation for L, we get

$$L = Z(V_\infty)^{2-e-f} \rho_\infty^{1-f} S^{1-f/2} a_\infty{}^{e} \mu_\infty{}^{f}$$

Rearranging the terms, we get,

$$L = Z\rho_\infty V_\infty{}^2 S \left(\frac{a_\infty}{V_\infty}\right)^e \left(\frac{\mu_\infty}{\rho_\infty V_\infty S^{1/2}}\right)^f$$

The term $a_\infty/V_\infty = 1/M_\infty$, where M_∞ is the free stream mach number.

The dimensions of S is l^2. Therefore the quantity $\sqrt{s}$ can represent the chord c.

The quantity $\mu_\infty/\rho_\infty V_\infty c$ can be represented by $1/R_e$, where R_e is the non-dimensional Reynolds number. The equation for lift L becomes

$$L = Z\rho_\infty V_\infty^2 S\left(\frac{1}{M_\infty}\right)^e\left(\frac{1}{\mathrm{Re}}\right)^f$$

We now obtain the value for lift coefficient as

$$c_l/2 = Z\left(\frac{1}{M_\infty}\right)^e\left(\frac{1}{\mathrm{Re}}\right)^f$$

We now write lift L as

$$\underset{\text{Lift}}{L} = \underset{\text{Dynamic pressure}}{q_\infty} \times \underset{\text{Wing area}}{S} \times \underset{\text{Lift coefficient}}{c_l}$$

The Reynolds number R_e is the **ratio of inertial forces to viscous forces** with in the fluid which is subjected to relative internal movements due to different fluid velocities caused by the formation of boundary layer.

Reynolds number $1/R_e = (\mu_\infty/\rho_\infty V_\infty c)$

By matching the dimensions of all terms on both sides and equating them, we arrive at the values of these dimensionless constants.

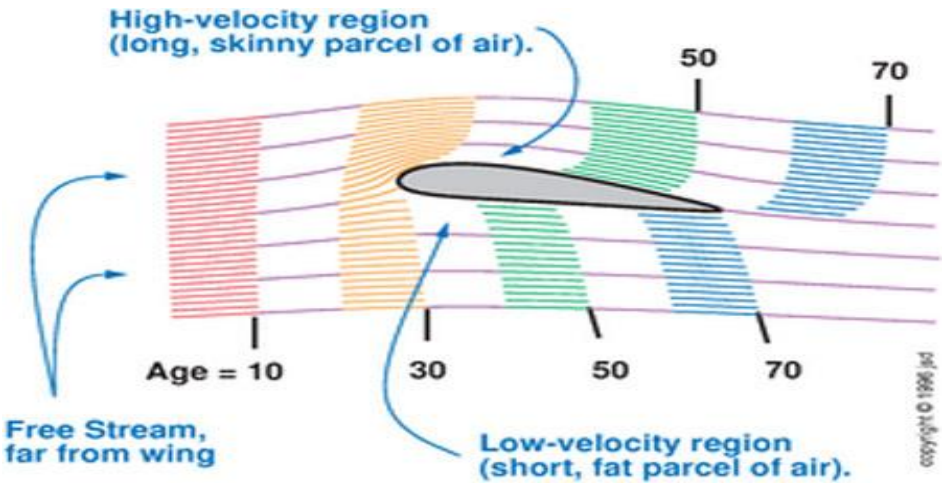
We can now define the lift force L as, $\mathbf{L = \frac{1}{2}\rho V^2 S\, C_l}$, where C_l is called the lift coefficient and is dependent on Reynolds number and Mach number.

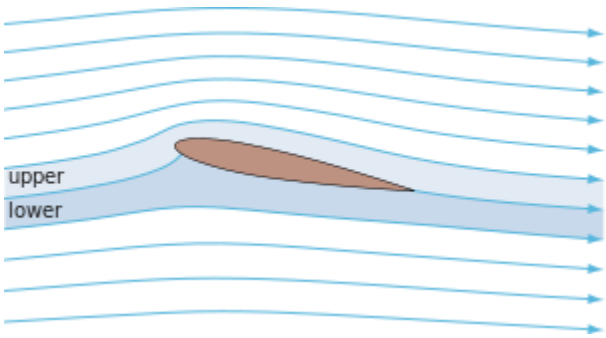
The quantity $\frac{1}{2}\rho V^2$ is the dynamic pressure.

Mechanism of Lift production: Lift is produced by the development of pressure differential over the airfoil. The generation of lower lift on top surface and higher pressure on the bottom surface is explained below:

Explanation for lift production using Bernoulli's equation and Euler's equations.

The flow velocity distribution over airfoil is shown below:





Consider the shaded stream-tubes upper and lower sides of the airfoil. The shaded upper stream-tube flowing over the top of the airfoil, gets squashed to a smaller cross section as it flows over the nose of airfoil after stagnation point. As it gets compressed over the top, the velocity must increase to satisfy the "mass continuity". As the stream-tube flows over the bottom face of the airfoil, the cross-sectional area increases, thereby causing reduced velocity.

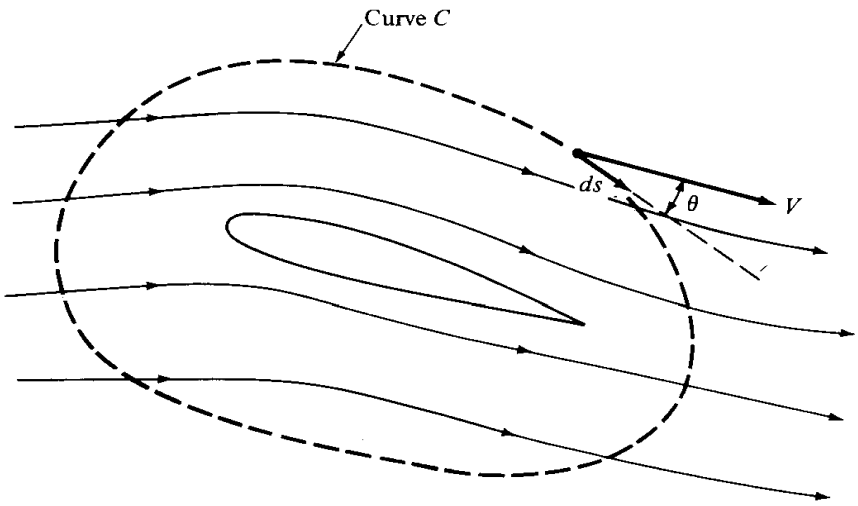
The velocity differential causes pressure differential keeping the flow energy, p + ½ ρV^2, constant. Also from the Euler's equation dp = $-\rho$V dV, when velocity increases (dV positive), the pressure decreases (dp negative).

Finally, it follows that owing to the lower pressure over the top surface and higher pressure on the bottom surface, the airfoil feels a lift force in the upward direction.

Generation of Lift using circulation theory:

The circulation theory is more of a mathematical formulation for calculation of lift for the airfoil. Consider the flow over the airfoil, with a closed curve C

drawn around the airfoil.



At a point on curve C, the flow velocity is V. Let the velocity vector make an angle of θ with the tangent to the curve. Let ds be an incremental distance along the curve.

Then, a quantity called "circulation Γ" is defined as

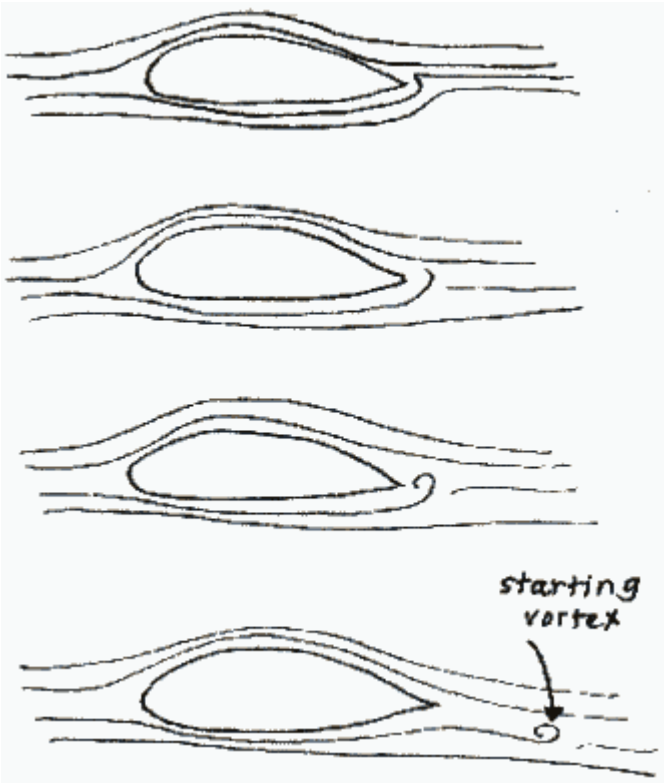
$$\Gamma \equiv \oint_C V \cos\theta \, ds$$

Where Γ is the line integral of the component of flow velocity along curve C. The lift per unit span is calculated from

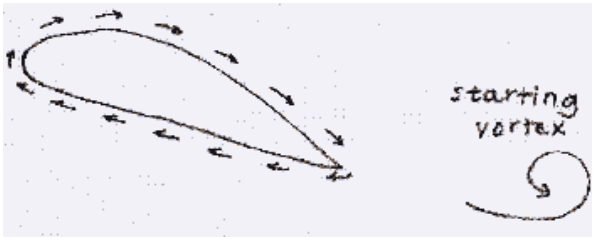
$$L = \rho_\infty V_\infty \Gamma_\infty$$

Circulation was explained by Kutta-Joukowski, as below:

Viscosity is essential in generating lift. The effects of viscosity lead to the formation of the starting vortex (see Figure 4), which, in turn is responsible for producing the proper conditions for lift.



The starting vortex rotates in a counter-clockwise direction. To satisfy the conservation of angular momentum, there must be an equivalent motion to oppose the vortex movement. This takes the form of circulation around the wing, as shown in Figure 5. The velocity vectors from this counter circulation add to the free flow velocity vectors, thus resulting in a higher velocity above the wing and a lower velocity below the wing.



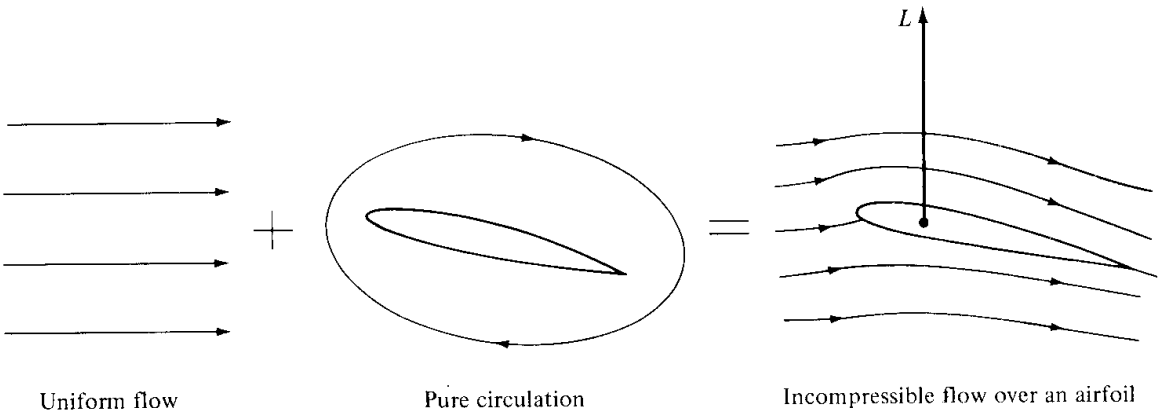
Fluid dynamics mathematicians have understood that in order for lift to be produced, a body must produce a finite amount of "**circulation**" (see **Kutta-Joukowski Theorem**).

The creation of a starting vortex released from trailing edge at a certain velocity and angle of attack, is called "Kutta condition".

A body with a sharp trailing edge which is moving through a fluid will create about itself a circulation of sufficient strength to hold the rear stagnation point at the trailing edge."

This statement usually holds for airfoils since most of them have pointy trailing edge.

Addition of two elementary flows around the airfoil produces the lift force L, directly proportional to the circulation Γ, as shown below:



Can an airfoil produce lift when it is flying upside down?

Yes. But the coefficient of lift will be different and less for the upside down airfoil as shown below:

Variation of Lift with angle of attack α:

The wings provide lift by creating a situation where the pressure above the wing is lower than the pressure below the wing. Since the pressure below the wing is higher than the pressure above the wing, there is a net force upwards.

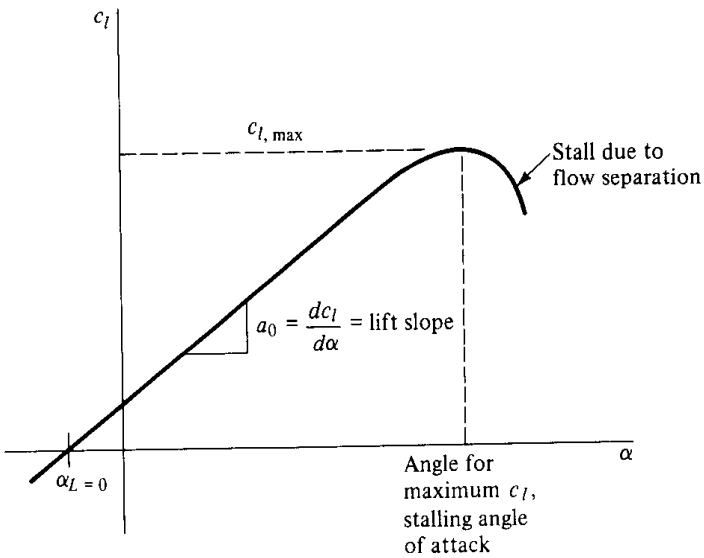
To create the pressure difference, the surface of the wing must satisfy one or both of the following conditions. The wing surface must be:

1. Cambered (curved); and/or
2. Inclined relative to the airflow direction

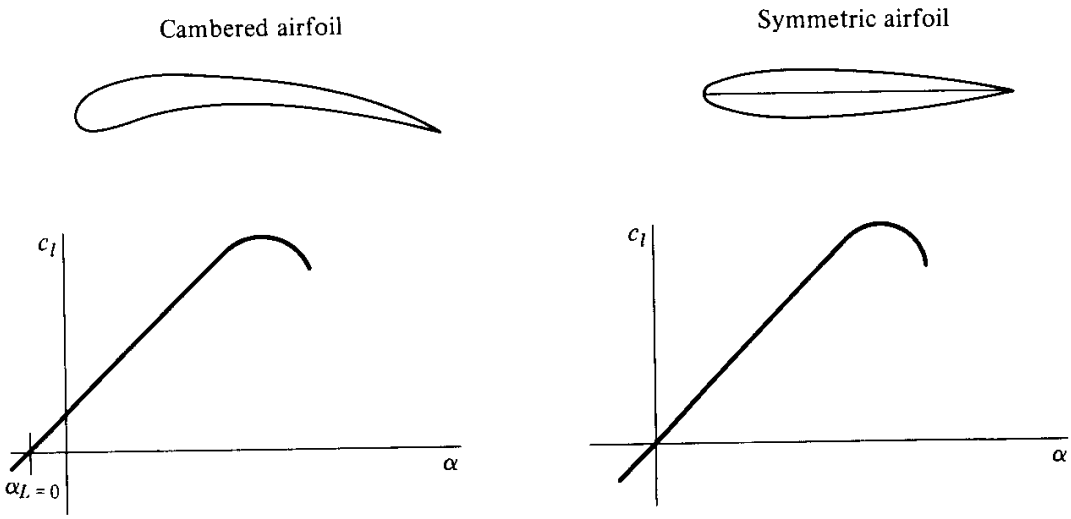
Lift variation indicates linear relationship with angle of attack α, as given below**:**

Lift Curve:

- Lift varies linearly over a large range of angle of attack variation
- There is a positive value of C_l when α = 0, due to the positive camber of the airfoil.
- It indicates that lift force is generated even at zero angle of attack
- All airfoils with such positive camber have to be pitched to some negative angle of attack before zero lift is obtained. The value of α when lift is zero is defined as zero lift angle of attack $\alpha_{L=0}$.
- At high values of α, lift drops suddenly, indicating "stalled condition'
- Stalled condition is caused by boundary layer separation on the surface or the airfoil.



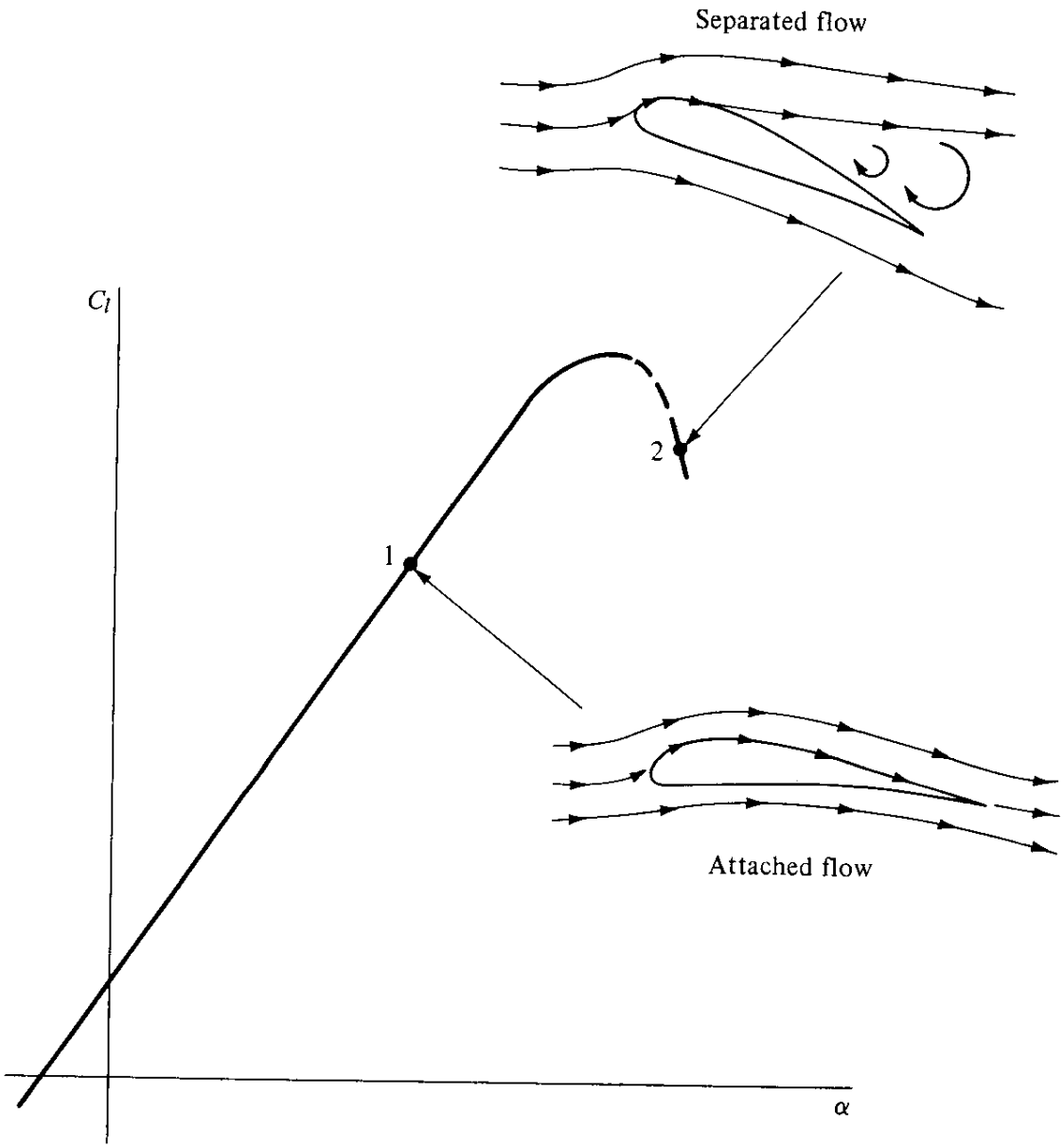
Comparision of lift curve for symmetric and cambered airfoils:



The lift curve for symmetric airfoil goes through origin indicating zero lift at zero angle of attack.

Flow mechanism associated with stalling:

The lift curve below indicates the flow separation at high angles of attack leading to sudden drop in lift and stalling of the airfoil.

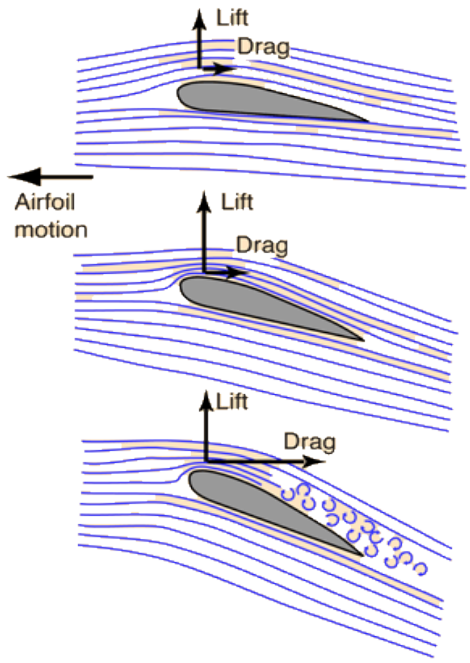


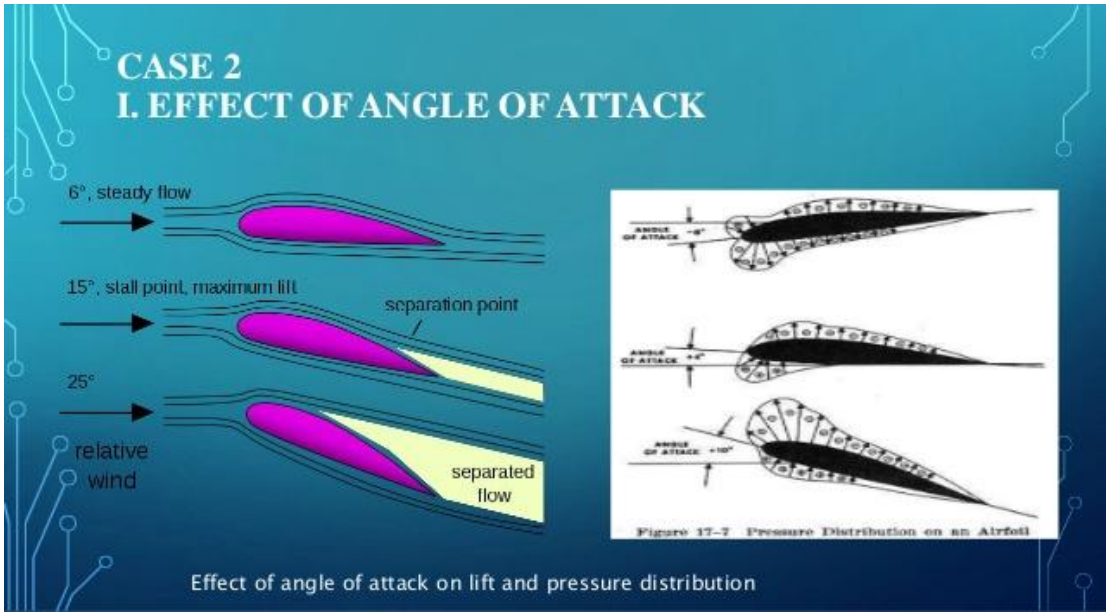
At point 1 on the linear portion of the lift curve, the flow field over the airfoil is attached to the surface. However, the effect of friction is to slow the airflow near the surface. There will be an adverse pressure gradient on the surface slowing down the flow.

As the angle of attack is increased, the adverse pressure gradient on the top surface will become stronger, and at some value of α, "the stalling angle of attack" the flow becomes separated from the top surface. When separation occurs, the lift decreases drastically and drag increases (pressure drag due to separation).

Flow patterns over airfoil at different angles of attack:

The flow patterns over a cambered airfoil as the angle of attack increases, is shown below:



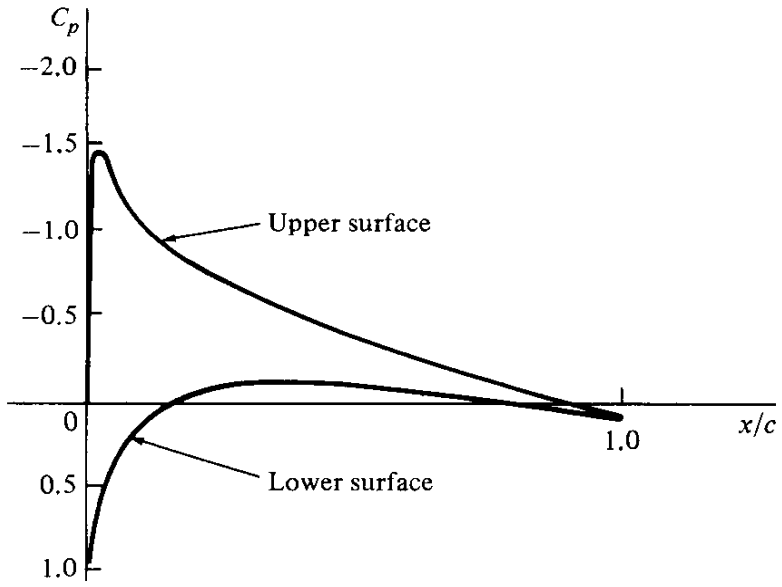


Pressure Coefficient: The pressure distributions on airfoil surfaces are usually given in terms of a dimensionless pressure coefficient C_p. Instead of plotting the actual pressure (in terms of units, say, newtons per square meter), the dimensionless pressure coefficient C_p, facilitates understanding the variation of aerodynamic forces with different parameters simple.

The pressure coefficient is defined as

$$C_p \equiv \frac{p - p_\infty}{q_\infty} \equiv \frac{p - p_\infty}{\frac{1}{2}\rho_\infty V_\infty^{\ 2}}$$

The distribution of C_p over top and bottom surfaces of NACA0012 airfoil at angle of attack of 3.9°, $M_\infty = 0.345$ and $R_e = 3.245 \times 10^6$, is given below:

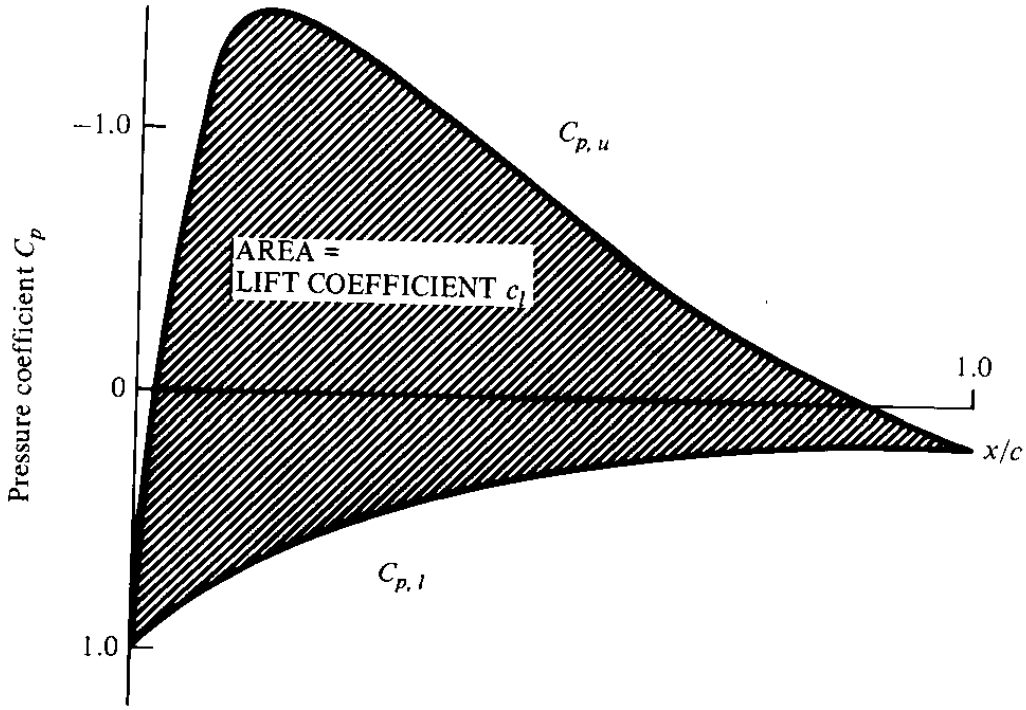


By convention, the plots of C_p over the airfoil is shown with negative values above the abscissa.

X axis represents the distance x, on top and bottom surfaces as fraction of chord c.

C_p at the leading edge is positive because $p > p_\infty$. However, as the flow expands over the top surface of the airfoil, p decreases rapidly and C_p goes negative as local pressure $p < p_\infty$.

The distribution of pressure coefficient over the airfoil leads directly to the value of lift coefficient. The values and distribution of pressure coefficient at different flow speeds and angles of attack can be found out experimentally in a wind tunnel.



Area of curves of pressure coefficient variation on upper and lower surfaces gives the lift coefficient.

Sources of Drag:

Drag is the force that resists the movement of a solid object through a fluid (a liquid or gas). Drag is made up of friction forces, which act in a direction parallel to the object's surface

It therefore acts to oppose the motion of the object.

Types of Drag

- Types of drag are generally divided into three categories: parasite drag, lift-induced drag and wave drag.
- Parasitic drag includes **form drag, skin friction drag and interference drag**.
- **Induced drag** is only relevant when wings or a lifting body are present, and is therefore usually discussed only in the aviation perspective of drag.

- **Wave drag** occurs when a solid object is moving through a fluid at or near the speed of sound in that fluid.

Form Drag or Pressure Drag: As the flow progresses over the surface, the pressure keeps increasing, causing a adverse pressure gradient. Boundary layer separates near the trailing edge. This imbalance of pressure forces causes pressure drag.

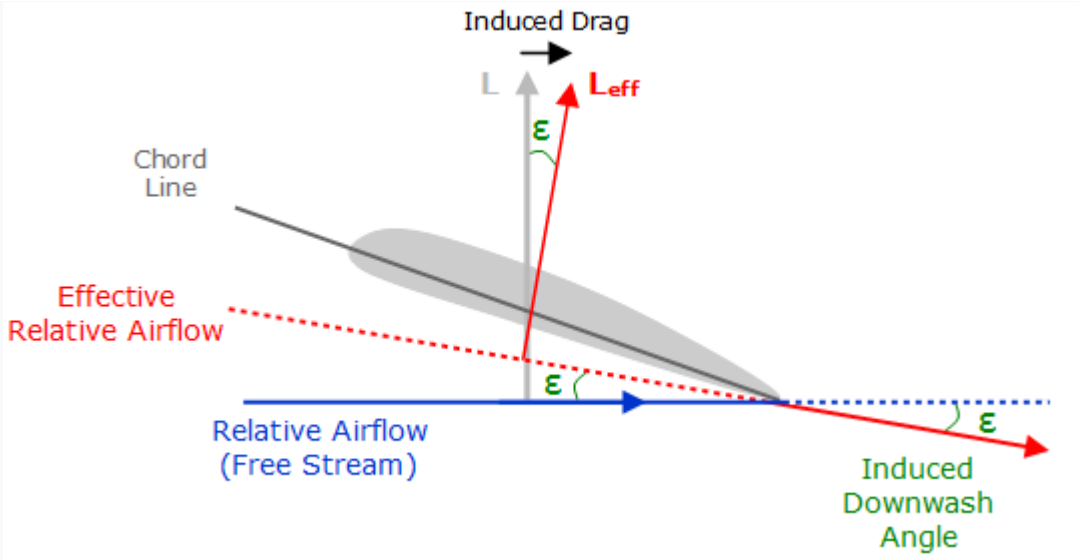
Skin friction drag: The presence of friction in the flow causes a shear stress at the surface of the body, which causes the skin friction drag.

Interference Drag: Flow resistance caused by one part of aircraft on another is termed as interference drag. This is maximum in case of interference between wings and fuselage.

Parasite Drag: Aircraft has many non-lift producing projections/protrusions like aerials, bomb/missile carrying pylons, sensors, dynamos etc. These parts do not contribute to lift, but result in additional drag, which is called parasite drag.

Induced drag: Airflow over the wing causes high pressure below the wing and low pressure above the wing. This causes air near the wing tips to flow from bottom surface to top of the wing, forming trailing vortices, a series of eddies behind the wing.

The trailing vortices result in induced downwash of airflow. The air behind the wing is deflected downwards. Thus the net direction of air which actually passes the airfoil is in a downward direction, so **the lift which is perpendicular to the airflow tilts backwards**. The component of tilted lift force parallel to the airflow direction contributes to the drag, called induced drag.



In the above figure, the vertical line labeled "L" is perpendicular to the free stream and indicates the orientation of the lift on the wing. The red vector labeled "L_{eff}" is perpendicular to the actual airflow in the vicinity of the wing; it represents the lift on the airfoil section in two–dimensional flow at the same angle of attack and results in induced drag.

$$M_{ac} = \text{constant}$$

Summary of Airfoil Drag: The total drag of an airfoil is the sum of three contributions,

$$D = D_f + D_p + D_W, \text{ where}$$

$$D = \text{Total drag on the airfoil}$$

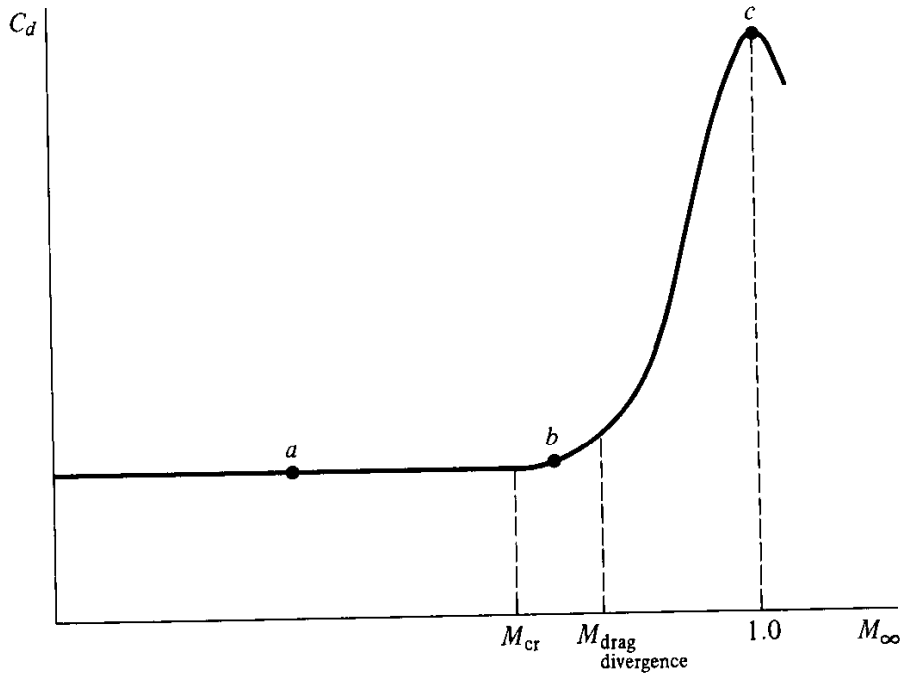
D_f = Skin friction drag

D_p = Pressure drag due to flow separation

D_w = Wave drag (present only at transonic and supersonic speeds, zero for sub-sonic speeds below drag divergence M ach number)

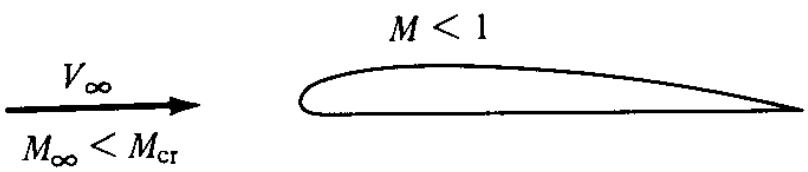
Variation of Coefficient of drag c_d with flight speed, Mach number:

Variation of c_d with mach number is shown below:

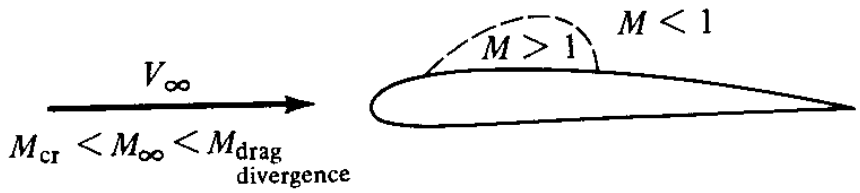


At low mach numbers, the airfoil drag coefficient is virtually constant, indicated by point a.

The flow over entire airfoil surface is subsonic during this phase as indicate below:



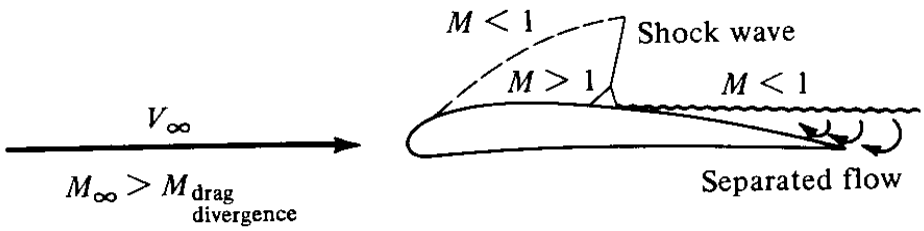
If M_∞ is increased slightly, a bubble of local supersonic flow will occur surrounding the minimum pressure point, as shown below. The Mach number at which the flow at any point on the surface of the airfoil becomes supersonic is termed as critical Mach number, M_{cr}.



The drag coefficient, c_d will remain reasonably low, as indicated by point b.

If M_∞ is further increased, a very sudden and dramatic rise in the drag coefficient will be observed, Here, shock waves will suddenly appear, causing sudden rise in the drag coefficient. Flow separation will also appear.

This raise in drag is attributed to wave drag. The free stream Mach number at which the drag coefficient begins to increase rapidly is known as "drag divergence Mach number".



(c)

Note that $M_{cr} < M_{drag\ divergence} < 1.0$.

The shock pattern associated with flight speeds between drag divergence mach number is called transonic region and will remain till free mach number of around 1.2. At M 1.2, the flow over all parts of airfoil will be supersonic.

Centre of Pressure: The centre of pressure is a point on the airfoil where all aerodynamic pressure field around the airfoil can be represented by a single force vector with zero moment.

The centre of pressure on a symmetric airfoil lies close to quarter chord point. It will not move as the angle of attack or lift coefficient changes.

However, on a cambered airfoil, the centre of pressure does not occupy a fixed point. At angle of attack corresponding to maximum lift coefficient, the centre

of pressure lies a little behind the quarter-chord point, but as lift coefficient reduces the centre of pressure moves to the rear.

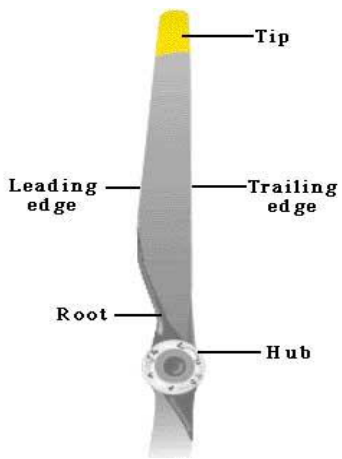
Rotary wing aircraft concepts-Forces while hovering, Propeller theory

A **rotary-wing aircraft** is a heavier-than-air flying machine that uses lift generated by wings, called rotor blades, that revolve around a mast.

A helicopter is a rotary wing aircraft, whose rotors are driven by the engine(s) throughout the flight to allow the helicopter to take off vertically, hover, fly forwards, backwards and laterally, as well as to land vertically. Helicopters have several different configurations of one or more main rotors.

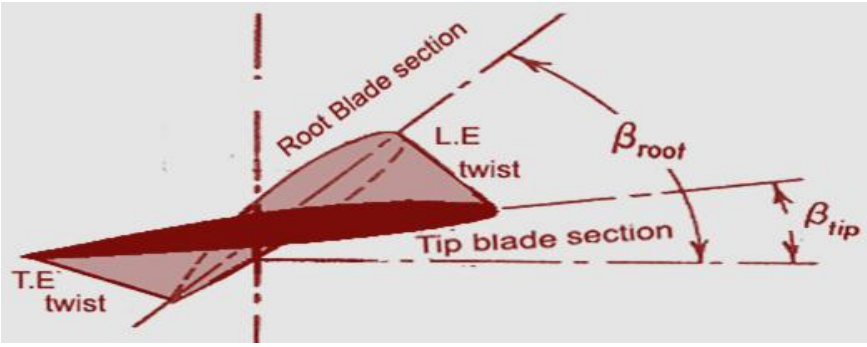
Helicopters with a single shaft-driven main lift rotor require some sort of anti-torque device such as a tail rotor to provide stability.

Propeller geometry:



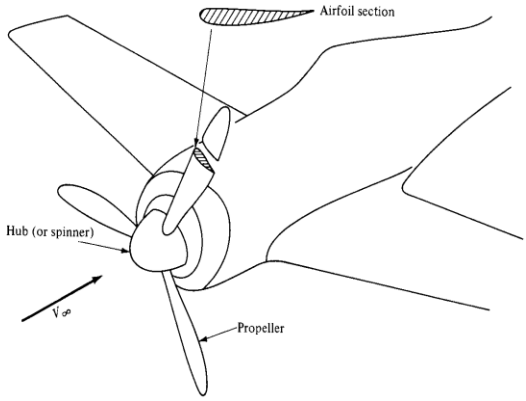
Propellers have two or more exactly similar blades, each of which is built by stacking of airfoil sections radially from root to tip.

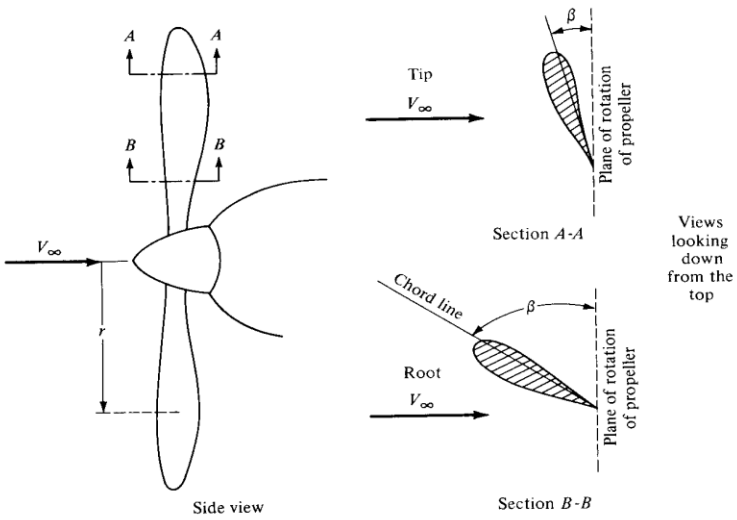
Blade Twisting: The blade shape has a radial twist which gives a different local angle of attack for each blade section, in order to provide uniform lift generation over the entire blade.



Blade Geometry:

- **Propeller Disc**: The propeller disc is the circular area traced out by the rotating propeller blades.
- **Blade Section**: Each section of the blade is an airfoil, usually highly cambered. The size, shape and local angle of attack (twist) may vary along the span of the blade.
- **Pitch angle**: The angle between the chord line and the propeller's plane of rotation is called Pitch angle β.
- **Pitch**: Pitch refers to the forward movement of the propeller for one revolution of the blade section.





Operation: A propeller is a group of rotating blades, or "wings", oriented to the direction of relative velocity of air stream, that the lift is produced in the forward direction.

The lift produced by each airfoil section is perpendicular to the effective resultant air velocity approaching the blade at that point. The resultant air velocity is the vector sum of the velocity due to the forward speed of the aircraft, the velocity due to the rotation of the aircraft and the induced downwash velocity at the blade tips.

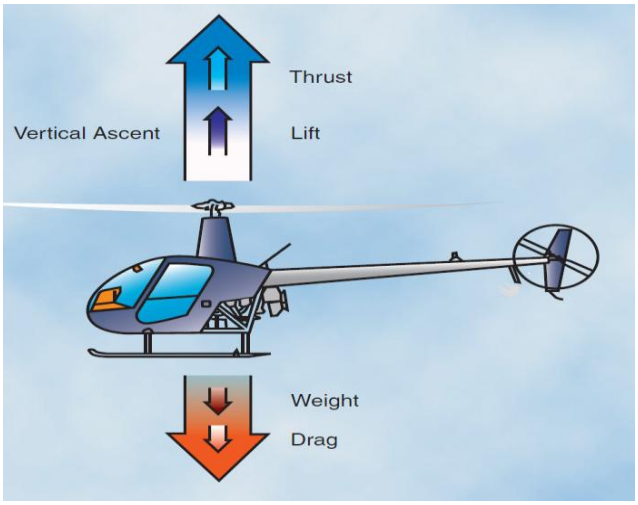
The lift vector is inclined at an angle to the flight direction and its component perpendicular to the flight direction

When a helicopter's rotor moves through air, it pushes air downwards through a combination of airfoil design, which moves air downwards. The upward angle to the blade relative to the oncoming air pushes the air to the underside of the rotor. By Newton's third law, the force pushing the air downwards, creates an equal and opposing force on the rotor. This force is the Lift overcoming the weight while hovering and also causing lateral/forward movement of helicopter when the rotational axis is tilted.

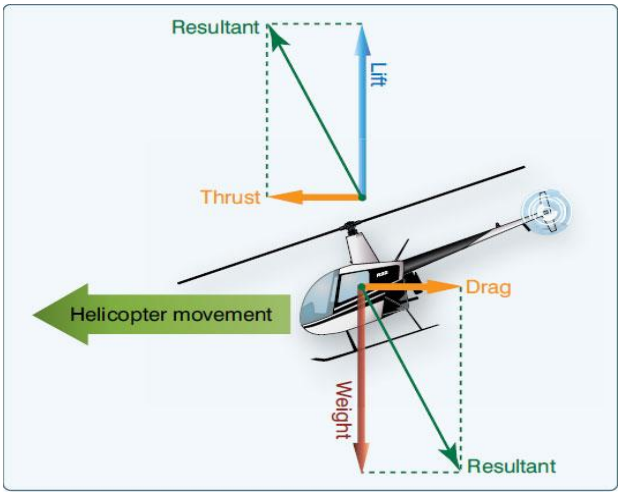
The tail rotor counters the torque generated by the rotating rotor.

Forces while hovering: During hovering flight, the tip path plane of the rotor is horizontal. Lift and thrust act straight up, weight and drag act straight down. The sum of lift and thrust forces must be equal to the sum of weight and drag forces during hovering condition.

If the sum of lift and thrust in a vertical hovering condition increases the sum of weight and drag forces, then the helicopter will rise vertically up.



Forces during forward-flight: For forward flight, the tip-path plane is tilted forward, thus tilting the total lift-thrust force forward from the vertical. This resultant lift-thrust force can be resolved into two components: lift acting vertically upward and thrust acting horizontally in the direction of flight. In addition to lift and thrust, there is weight, the downward acting force, and drag, the rearward acting or retarding force of inertia and wind resistance.



In straight-and-level, un-accelerated forward flight, lift equals weight and thrust equals drag. (Straight-and-level flight is flight with a constant heading and at a constant altitude.) If lift exceeds weight, the helicopter climbs; if lift is less than weight, the helicopter descends. If thrust exceeds drag, the helicopter increases speed; if thrust is less than drag, it decreases speed.

Propeller Theory: Propeller theories are used to design and predict the propeller performance using the fundamental aerodynamic/fluid flow dynamic concepts. There are two theories in use. They are:

- Blade Element Theory
- Momentum Theory

Blade Element Theory: Blade element theory use propeller blade shapes made up of stacked airfoils. In practice a large number of different airfoils are used to make up one propeller blade.

The blade elements are assumed to be made up of airfoil shapes of known lift, C_l and C_d characteristics.

Each of these elements shall have its own lift, C_l, drag, C_d coefficient characteristics.

The thrust, dT, created by an blade element will contribute to the elemental lift dL, and drag dD.

Propeller thrust and torque are now computed by integrating from root to tip of the blade and for the total number of blades. Thus the net thrust and torque are seen to be directly proportional to the number of blades, and the shape (chord/camber).

Momentum Theory: Momentum theory does not use the blade shape for the design and analysis of propellers.

The propeller is assumed to be replaced by an '*actuator disk*', a flow energizer.
The 'disk' is assumed to be of very small thickness and is a continuous and 100% porous body of no mass, with a projected frontal area '**A**' (*swept area*) equal to the annulus of the rotating propeller blades.
There is no 'resistance' (i.e. drag) of the air passing through the 'actuator disk', (since there are no propeller blades)

The momentum analysis shows that the air flow velocity through the actuator disk is the mean of the velocities far upstream and far downstream of the propeller

This conclusion is drawn from a simplified model, where the thrust force

$$T = \frac{1}{2} \rho \left(V_e^2 - V_a^2 \right) A$$

UNIT - III

Propulsion- Aircrafts, Rockets and Missiles: Thrust for flight, Reciprocating engines-2 stroke/4 stroke; Jet engine and types, Rocket engines - Description, Principles of operation. Types of Missiles, similarities and differences with launch vehicle, controls for missiles, Airframe components of missiles

Propulsion-Aircraft, Rockets and Missiles
Thrust for flight; Reciprocating engines-two-stroke & four-stroke; the jet engine, rocket engines-description, types, principles of operation.
Types of missiles; similarities & differences with launch vehicles;
Control for missiles, airframe components of missiles

Thrust for Flight- Reciprocating Engines; Jet Engines:

Types of Aerospace Propulsion:

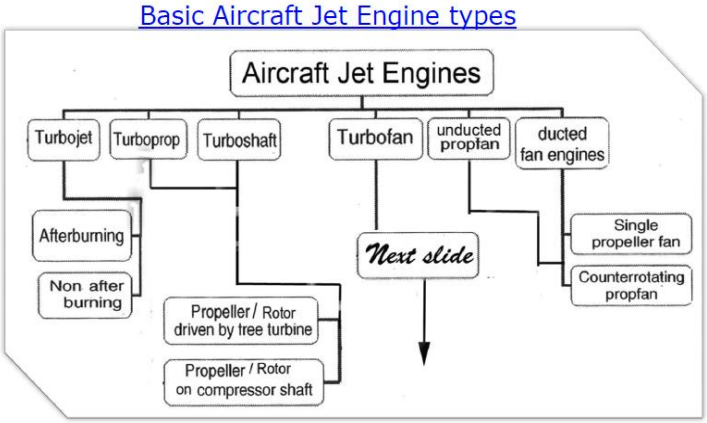
Air Breathing Systems: Broadly grouped as - Reciprocating& Jet Propulsion Engines.

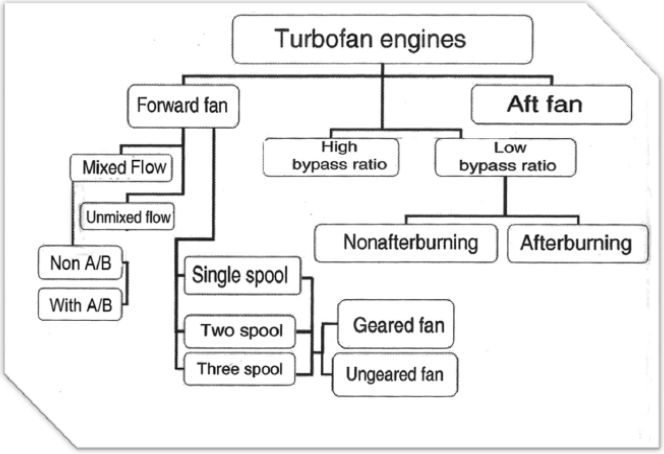
- Reciprocating Engines
- Gas Turbine Engines
- Ram Jets, Pulse Jets & Scram Jets

Non Air Breathing Systems

- Rockets

Basic Aircraft Jet Engine types





1.2.2. Working Principles, Advantages/Disadvantages & Applications:

1.2.2.1: Reciprocating Engines (I.C Engine): Working Principle: The four strokes of anInternal Combustion (I.C)engine are Intake, Compression, Power and Exhaust strokes.

During intake stroke, the piston moves downwards and the mixture of fuel and air (charge) is admitted in to the cylinder. At the completion of intake stroke, the inlet valve closes.

During the compression stroke, the piston moves up, compressing the charge. At the end of compression stroke, the electric spark ignites the charge.

On ignition, combustion of air fuel mixture releases thermal energy, exerting high force on the piston. This commences the power stroke.

During the power stroke, the piston is driven downwards.

Once the power stroke is completed, the exhaust valve opens. While the piston is moving up, the combustion gases are driven out of the cylinder through the exhaust valve. This creates a suction in the cylinder, that initiates the next cycle of operations.

The reciprocating movement of piston is transmitted to the crankshaft and converted into rotary motion. The crankshaft is connected to the propeller, which produces the forward thrust force for the aircraft.

The rotating output shaft of the I.C engine can be connected to a propeller, ducted fan, or helicopter rotor.

The propeller displaces a large mass of air rearwards, accelerating it in the process.

Reciprocating engines can produce up to 4000 KW power. Power to weight ratio (P/W) of up to 1.4 is produced.

The power produced by an I.C engine is given by

P = $\frac{KNV_c\rho_{air}fQ_f\eta_o}{60}$ where

K = constant; either 1.0 for 2 stroke engine or 0.5 for 4 stroke engine

N = rpm (around 5000-9000 rpm)

V_c = Volume of the cylinder

ρ_{air} = density of air

f = fuel air ratio (usually 13 to 15 ieone part fuel to 15 parts of air to burn the fuel completely)

Q_f = Calorific value of fuel (kerosene- 42 MJ/kg)

η_o = overall efficiency (usually 0.25 to 0.35)

$KNV_c\rho_{air}$ is the mass flow rate ingested in to the engine

- Multiplying mass flow rate with f gives the amount of fuel
- Multiplying with Q_f gives the heat energy released

To increase the power of the I.C engine, we need to

- Increase N –increases P
- As altitude increases ρ decreases, and P reduces. To offset this, turbo superchargers are used.

Advantages of Reciprocating Engines:

- Reciprocating engines provide excellent fuel economy and good take-off characteristics within their range of operations

- Highly suitable for small aircraft flying up to 500 km/hr and operating at low altitudes
- Components of reciprocating engines are subjected less thermal stresses than gas turbine-propeller combination
- Aircraft fitted with reciprocating engines need short runways
- Mainly used for business travel, farming & agriculture, air-taxi/ambulance, pilot training and unmanned aerial vehicles

Disadvantages of Reciprocating Engines:

- Reciprocating engines suffer drop in power at altitudes
- Difficulty in cooling and lubrication
- Low Power/Weight ratios compared to gas turbine engines
- Need high octane fuels to improve power output
- Increase in power output require larger number of cylinders, thereby increasing the frontal area and weight
- Use of reciprocating engines is limited to low speeds and altitudes
- Development reached a saturation stage as far as maximum power is concerned
- Maintenance requirement of piston-prop engines is more than turbojet aircraft
- Exhaust gases have less impurities in turbojet engines

1.2.2.2: Aircraft gas turbine Engines

All modern aircraft are fitted with gas turbine engines. Gas turbine engines can be classified into the following:-

(a) Turbojet engines

(b) Turbofan engines

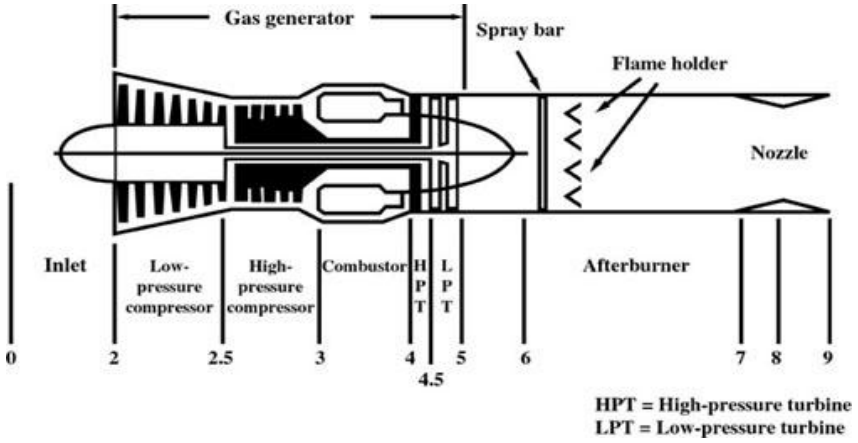
(c) Turbo-shaft engines

(d) Turboprop engines

Taken in the above order they provide propulsive jets of increasing mass flow and decreasing jet velocity. Therefore, in that order, it will be seen that the turbojet engines can be used for highest cruising speed whereas the turboprop engine will be useful for the lower cruising speed at low altitudes.

In practice the choice of power plant will depend on the required cruising speed, desired range of the aircraft and maximum rate of climb.

Turbojet Engine: Schematic diagram of a turbojet engine with station numbering is given below:



Working Principle:

1. The thrust of a turbojet engine is developed by compressing the free stream air in the diffuser or inlet and compressor. The diffuser converts the kinetic energy of the entering air into pressure rise which is achieved by ram effect. Diffusion in the inlet occurs due the geometric shaping of the inlet.
2. The compressor is driven by the turbine. It rotates at high speed, adding energy to the airflow and at the same time squeezing (compressing) it into a smaller space. Compressing the air increases its pressure and temperature
3. Compressor types used in turbojets were typically axial or centrifugal.
4. Use of axial flow compressors enable high pressure ratios. Modern axial compressors are split into low pressure and high pressure spools, driven by corresponding two stage turbine. High compressor ratios of 15:1 or more can be achieved while improving stability of operation at off-design conditions.The high pressure air is then mixed with fuel and burnt in the combustion chamber under constant pressure condition.
5. The combustion gasses at high temperature and pressure are expanded in the turbine and the exhaust nozzle. The expansion of gasses in the turbine provides power to drive the compressor while the exhaust nozzle expands the gasses to atmospheric pressure, thereby producing propulsive force, thrust.
6. The net thrust delivered by the engine is the result of converting internal energy to kinetic energy.
7. The exhaust products downstream of the turbine still contain adequate amount of oxygen. Additional thrust augmentation can be achieved by providing an afterburner in the jet pipe in which additional amounts of fuel can be burnt.
8. Turbojet engines are most suitable for speeds above 800 km/hr and up to 3.0 mach number

Advantages of Turbojet:

1. Power to Weight ratio is about 4 times that of Piston-Prop combination
2. Simple, easy to maintain, requires lower lubricating oil consumption. Complete absence of liquid cooling reduces frontal area
3. Turbojets allow faster supersonic speeds up to 3.0 M
4. There is no limit to power output while piston engines reached their peak power, beyond which any increase will result in high complexity and greater weight/frontal area.
5. Speed of turbojet is not limited by the propeller.
6. Turbojets can attain higher speeds than turboprop aircraft

Disadvantages-Turbojet:

- Fuel economy at low operational speeds is very poor
- It has low take-off thrust and hence poor starting characteristics
- High operating temperatures and engine parts are subjected to thermal stresses

Application:Turbojet engine is highly suited for aircraft at speeds above 800 km/hr.

Advantages of Gas turbines over Reciprocating Engines:

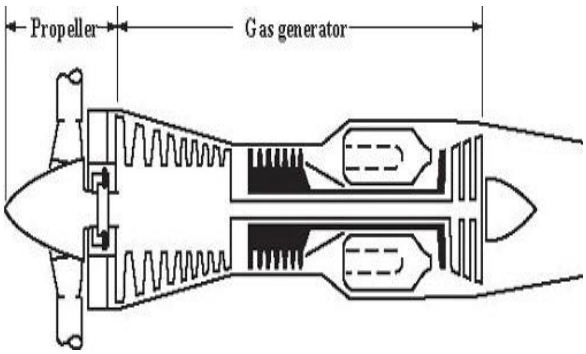
- **Mechanical Efficiency**: Mechanical efficiency of gas turbine engines is higher than reciprocating engines. This is mainly due to high friction losses in reciprocating engines.
- **Balancing**: Due to absence of reciprocating mass in gas turbine engines, balancing can be near perfect. Torsional vibrations are absent because gas turbine is a flow machine.
- **Smooth & Vibration-free operation**: Turboprop engines have fewer moving parts than piston-prop engines, offering greater reliability and time-between-overhaul (TBO).
- **Power**: The higher power of a turbo-prop engine allows it to fly at higher speeds and altitudes.
- **Shape**: Gas turbine engines have streamlined shape suitable from aerodynamic point of view.
- **Fuel**: Aviation turbine fuel is much cheaper than the high octane fuels used by reciprocating engines.
- **Lower Cost**: For a given power, gas turbine engine has lower cost and can be built faster
- **Weight:** Gas turbine engineshave higher power-to-weight ratios. This means, for a given weight, gas turbine engines develop more power.
- **Lubrication:** Lubrication in gas turbine engines is much simpler than reciprocating engines. The requirement is chiefly to lubricate the main bearing, compressor shaft and bearing auxiliaries.

- **High operational speed:** Turbine canbe run at much higher speed than reciprocating engine. Turbine can also be made lighter than the reciprocating engine of similar output. Therefore, for a given output, and higher speed, the torque can be lower. Gas turbine engines have better torque characteristics.
- **Silent Operation:** Since exhaust from gas turbine engines occurs under practically constant pressure conditions unlike the pulsating nature of the reciprocating engine exhaust, the usual vibrational noises will be absent in gas turbine engines.
- **Maintenance:** Relatively simpler in case of gas turbine engines.

Advantages of Reciprocating Engines over Gas turbine Engines:

- **Efficiency**: The overall efficiency of gas turbine engines is much less than the reciprocating engines.
- **Temperature Limitation**: The turbine blades in gas turbine engine are exposed to high temperature gasses continuously, and hence cannot exceed 1500 K.
- **Cooling**: We can achieve very good results by cooling the cylinder walls effectively. Cooling of turbine blades is complicated.
- **Ease of Starting**: It is more difficult to start a gas turbine than aa reciprocating engine.
- **Complexity**: Reciprocating engines are far less complex than their turbo-prop counter parts, from engineering considerations. This is primarily because of the high temperatures and forces unique to turbo-prop engine operation, which must be accommodated from materials and engine design.

1.2.2.3: Turboprop Engine: Schematic diagram is given below:



Working Principle: Turboprop engine is an intermediate between a pure jet engine and a propeller engine.

Turboprop engine provides high thrust per unit mass flow of fuel burnt by increasing mass flow of air. It offers better fuel economy. The propeller displaces a large mass of air rearwards, thereby increasing the net thrust.

The turbine extracts more power from the combustion gasses to drive the propeller. A small remaining energy is extracted by expansion in the jet nozzle.

The propeller and the compressor may be mounted on a single shaft or on separate shafts with a free turbine driving the propeller.

Advantages:

Turboprop engines have a higher thrust at take-off and better fuel economy.

The engine can operate economically over a wide range of speeds ranging from low speeds, where turbojet is uneconomical, to high speeds of about 800 km/hr where piston-prop engine cannot operate efficiently

It is easy to maintain and has lower vibration levels than piston-prop engine. The frontal area is much less than corresponding piston-prop engine.

Disadvantages:

The main disadvantage is that the propeller efficiency decreases greatly at high speeds due flow separation and shocks. The maximum speed is thus limited.

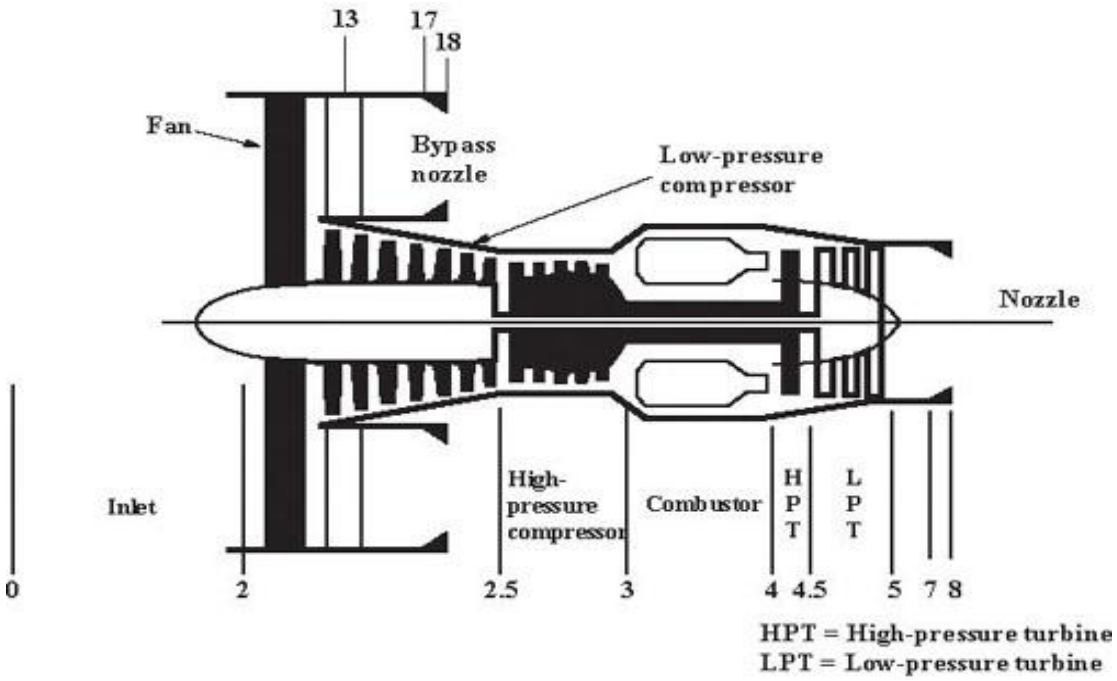
The turbine speeds need to be reduced through a suitable reduction gearing so that propeller runs at lower speeds, which adds to weight.

Applications:

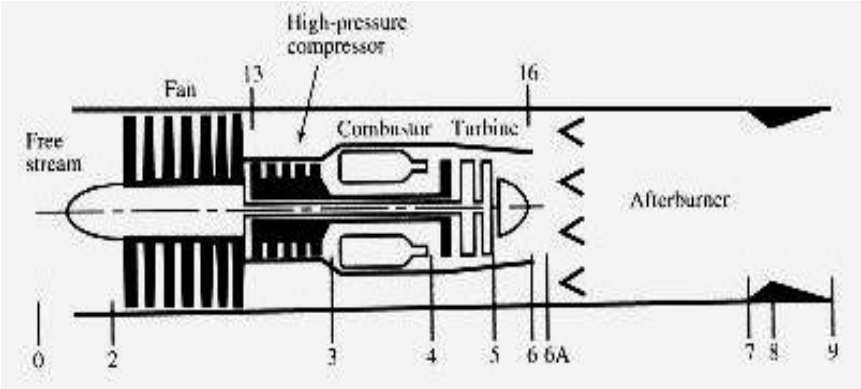
The turboprop engine is widely used in commercial and military aircraft due to its flexibility of operation and good fuel economy.

Turbofan Engine

Schematic Diagram of Turbofan (with station numbering): High by-pass ratio (used for commercial aircraft)



Turbofan with afterburner & Mixed flow: Low by-pass ratio (used for military aircraft)



Turbofan engine is designed as a compromise betweenturbojet and turbopropengines. The turbofan engine consists of a fan larger in diameter than the compressor, driven by the turbine. The fan displaces/bypasses free stream air around the primary engine. Two streams of air flow through the engine, primary airstream pass through the compressor and is delivered to the combustion chamber at high pressure to mix with fuel, while the other stream bypasses the primary engine to be expanded in the nozzle as a cold stream. The **hot and cold streams may be expanded through separate nozzles or combined together through a single nozzle**. The ratio of mass of cold air to the hot air is the by-pass ratio.

Thus the turbofan accelerates a larger mass of air at lower velocity than turbojet for a higher propulsive efficiency. Turbofan engines can also employ afterburner for higher thrust.

Turbofan engines can be aft-fan or forward fan (position of the fan), mixed or unmixed(hot and cold air streams) and high and low bypass ratio configuration

Advantages:

Fan is not as large as the propeller, therefore higher aircraft speeds can be attained without facing flow separation problems.

Turbofan engines do not encounter vibration problems associated with propellers. The fan could be encased in a duct/cowling to provide better aerodynamic shape.

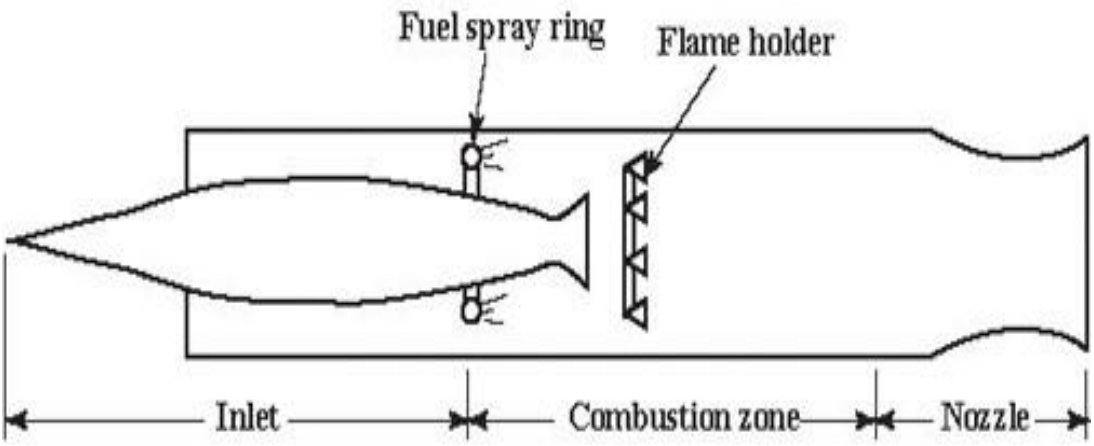
A **geared fan** connected to turbine reduces power consumed by the fan. It also produces low sound.

Turbofan is fuel efficient than turbojet, offers better propulsive efficiencies.

Lowers the sound levels of the exhaust gasses

Ramjet Engine

Schematic Diagram:



Operating Principle:

Ramjet Engine consists of supersonic diffuser, subsonic diffuser, combustion chamber and nozzle section

Air from atmosphere enters the supersonic diffuser at a very high speed. The air velocity gets reduced in the supersonic diffuser through normal and oblique shock waves.

Air velocity is furthur reduced in the subsonic diffuser.

The diffuser converts the kinetic energy of the entering air into static pressure and temperature rise which is achieved by ram effect. Diffusion in the inlet also occurs due the geometric shaping of the diffuser. The diffuser thus slows down the air enabling combustion.

Fuel is injected into the combustor through suitable injectors causing mixing of fuel with the air and the mixture is burnt

Combustion gases attain a temperature of around 1500-2000 k by continuous combustion of fuel air mixture

Fresh air supply continuously will not allow gasses towards the diffuser. Instead, gases are made to expand towards the tail pipe and nozzle, which expands the gases completely.

The gases leave the engine with a speed much higher than the air entering the engine.

The rate of increase of momentum of the working fluid produces thrust F in the direction of flight

Distinguishing Features:

Air enters the engine at supersonic speeds, must be slowed down to subsonic value, to prevent blow out of the flame in the combustor

Velocity must be low enough (approximately around 0.2-0.4 mach number) to allow mixing of fuel and stable combustion

Cycle pressure ratio depends on the diffusion pressure ratio. Very high pressure ratios of about 8 to 10 through ram compression is possible, therefore, a mechanical compressor is not required

Slowing down speeds from mach 3.0 to 0.3 will result in a pressure ratio of more than 30

As the ram pressure increases, a condition is reached where the nozzle gets choked. Thereafter, the nozzle operates at Mach 1 condition at throat

Advantages:

Ramjet is a simple machine and does not have any moving parts

Since turbine is not used, maximum temperature allowed is very high, around 2000 C, as compared to around 900 C in turbojets.

We can burn air/fuel ratios of 13:1which gives greater thrust

Specific fuel consumption is much better than other gas turbine engines, at high speeds and altitudes

Wide range of fuels can be used

It is very cheap to produce; adoptable for mass production

It is not possible to start a ramjet engine without an external launching device

The engine heavily relies on the diffuser and it is very difficult to design a diffuser which gives good pressure recovery over a wide range of speeds

Due to high air speed, the combustion chamber requires flame holders to stabilize the combustion

At very high temperatures of about 2000 C, dissociation of combustion products take place, reducing the efficiency of the plant

High fuel consumption at low speeds

Applications:

Highly suitable for propelling missiles.

Used in high speed military aircraft, in a combined cycle engine (Turbojet-Ramjet combination).

Development is in progress for a hypersonic aircraft system using turbojet-ram-scramjet combined cycle.

Subsonic ramjets are used as target weapons in conjunction with turbojet aircraft.

Pulsejet Engine:

Schematic Diagram:

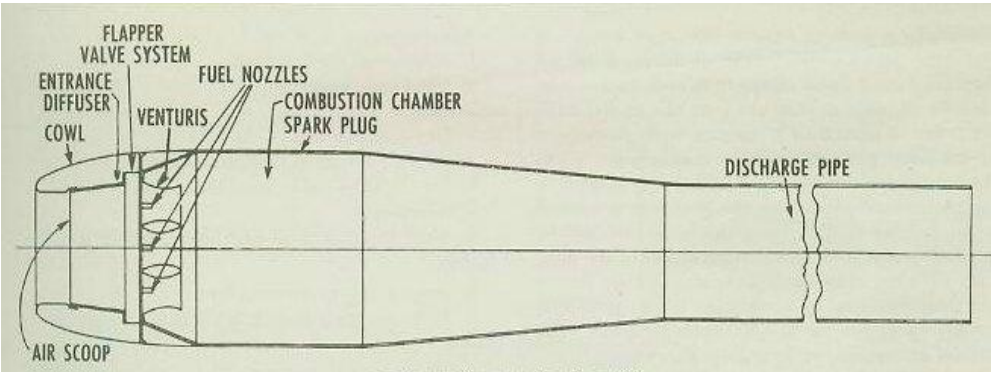


FIGURE 11D3.—A pulse-jet engine.

Basic Components are diffuser, Valve grid with spring loaded flapper valves, Combustion chamber with spark plug, tail pipe and discharge nozzle

Operation:

The diffuser converts the kinetic energy of the entering air into static pressure rise and slows down the air. Ram action also builds pressure in the diffuser.

The pressure differential opens the flapper valves which are spring loaded and the high pressure air enters the combustion chamber.

Fuel is injected and ignited by the spark plug

Combustion proceeds at constant volume with sudden explosion.

There is a sudden pressure rises in the combustion chamber which closes the flapper valves

The combustion gasses expand in the nozzle and escape to the atmosphere at high velocity

As combustion products leave the combustion chamber, a low pressure is created which causes the valves to open and a new charge of air enters the chamber

Distinguishing feature: Since the combustion chamber builds pressure, the engine can operate in static condition also. Proper design makes the duct to fire at a given pulse rate which can be as high as 500 cycles/sec

Advantages:

1. Simple to construct and hence cheap.
2. Can be mass produced in a short time.
3. Since it does not have any moving parts like compressor of turbine, it can be used in high temperatures.
4. Can be used for military applications.

Disadvantages:

1. It is having limited flight speed only.
2. Limited flying altitude.
3. High vibration and noise due to the pulses of thrust produced

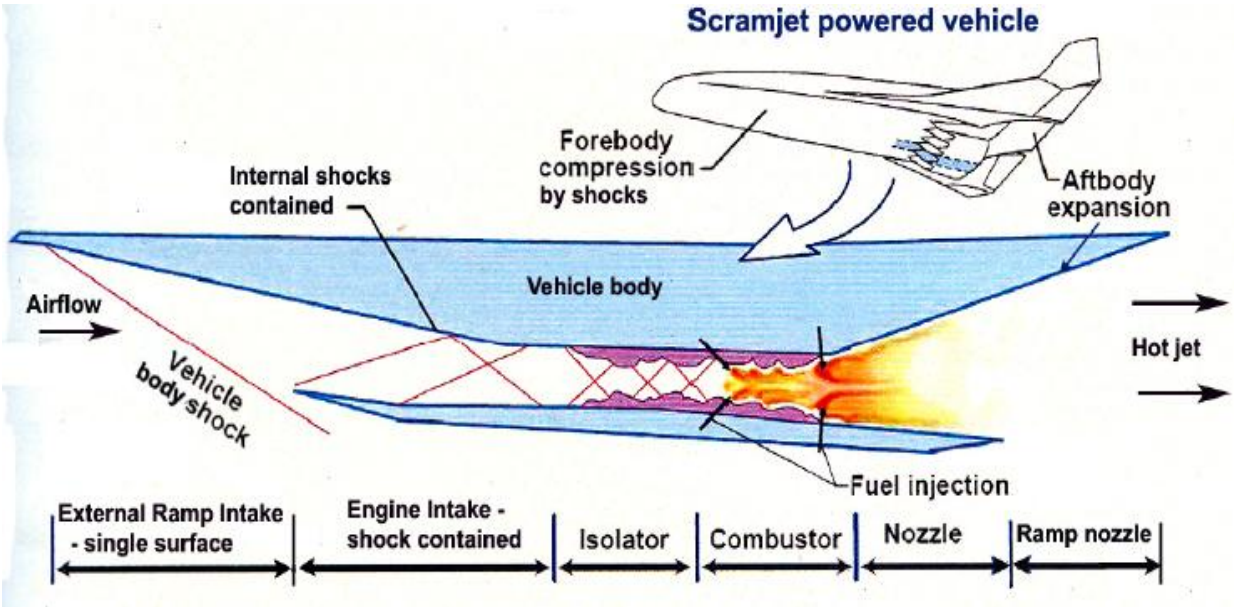
Scramjet Engine:

- Scramjet engine stands for supersonic combustion ramjet engine.
- The flow speed in the combustion chamber is supersonic
- Scramjet engine is characterized by high flow speeds ie low residence times in the engine.
- The engine needs larger combustion volumes; leading to integrated design of airframe and engine.

- In scramjet aircraft, the entire lower body of the aircraft is occupied by the engine. The front portion of the underside operates as external/internal diffuser, with rear portion providing expansion surface.

The scramjet consists of

- Diffuser (compression component) consisting of external ramp intake and engine intake
- Isolator
- Supersonic combustor
- Exhaust nozzle or aft body expansion component



Scramjet Engine- Construction:Scramjet engine is characterized by slow reaction times and high flow speeds ie low residence times in the engine. The engine needs larger combustion volumes; leading to integrated design of airframe and engine. In scramjet aircraft, the entire lower body of the aircraft is occupied by the engine. The front (fore) portion of the underside operates as external/internal diffuser, with rear (aft) portion providing expansion surface.

The scramjet consists of

- Diffuser (compression component) consisting of external ramp intake and engine intake
- Isolator
- Supersonic combustor

- Exhaust nozzle or aft body expansion component

Diffuser

- It consists of fore-body external intake and internal intake
- The fore-body provides the initial external compression and contributes to the drag and moments of the vehicle.
- The internal inlet compression provides the final compression of the propulsion cycle.

Since the flow upstream is supersonic, the geometry of the diffuser is entirely convergent.

Isolator: Isolator is constant area diffuser containing the internal shock structure, swallowed during supercritical operation of the inlet (or during operation after the inlet "started"). The isolatoris inserted before thecombustor to diffuse the flow further, through a shock train, producing desired flow speeds in the combustors. The function of the isolator is:

- The shock train provides a mechanism for the supersonic flow to adjust to a static back pressure higher than its inlet static pressure
- The isolator cross-sectional area may be constant or slightly divergent to accommodate boundary layer separation.
- When the combustion process begins to separate the boundary layer in the combustor, a pre-combustion shock train forms.
- The shock structure allows the required pressure rise, thus isolating the combustion process from the inlet compression process. Thus the isolator functions to prevent inlet surge or "unstart".

Combustor: Main features include:

- Avoidance of hot pockets near the walls implies that the fuel be injected from centrally located struts.

- The usual circular configuration for combustors can be sacrificed in favor of a rectangular configuration.

- Typical velocities in the combustion chamber are about 1 to 1.5 km/s and the Mach numbers will be 1.4 to 2.3 for a typical combustor entry Mach number of 2.5

Combustion limits: Two limits are very critical for the operation

- First, since when a supersonic flow is compressed, it slows down, the level of compression must be low enough (or the initial speed high enough) not to slow the gas below Mach 1. If the gas within a scramjet goes below Mach 1 the engine will "choke", transitioning to subsonic flow in the combustion chamber. Additionally, the sudden increase in pressure and temperature in the engine can lead to an acceleration of the combustion, leading to the combustion chamber exploding.
- Second, the heating of the gas by combustion causes the speed of sound in the gas to increase (through increase of $\sqrt{t}$and hence cause Mach number to decrease) even though the gas is still travelling at the same speed. Forcing the speed of air flow in the combustion chamber under Mach 1 in this way is called "**thermal choking**".
- A thermal throat results when the flow is slowed through tailored heat for causing dual-mode operation.
- There are engine designs where a ramjet transforms into a scramjet over the Mach 3-6 range, known as dual-mode scramjets.

Expansion System:

- The expansion system, consists of
  a. Internal nozzle

  b. Vehicle aft body
- It completes the propulsion flow path and controls the expansion of the high pressure and temperature gas mixture to produce net thrust.

Applications of Scramjets:

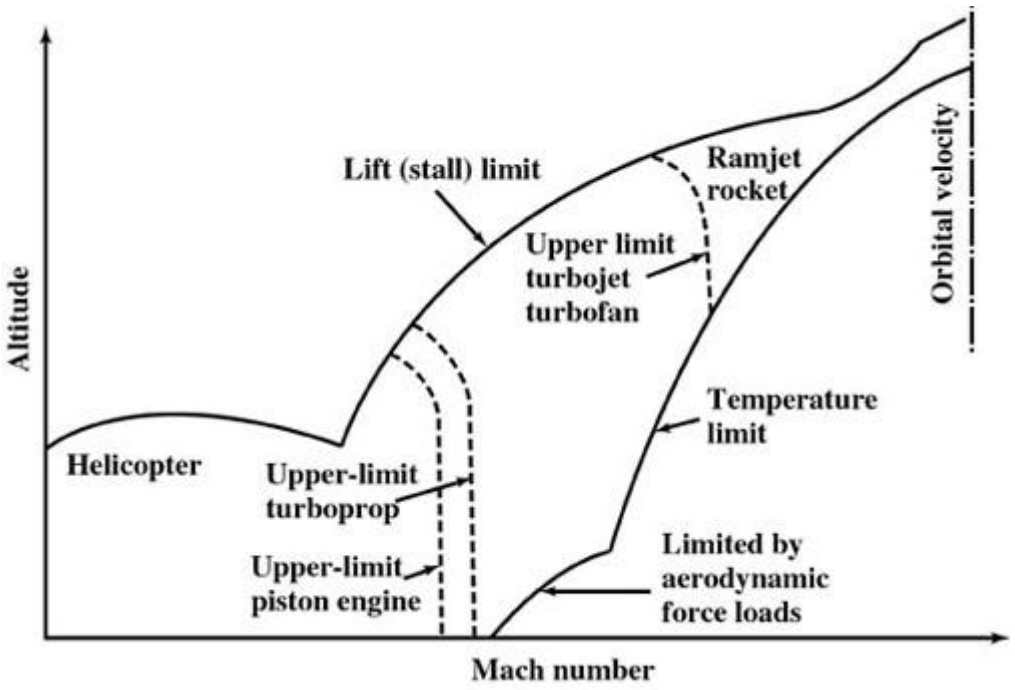
- Weapons systems -hypersonic cruise missiles
- Aircraft systems - global strike / reconnaissance
- Space access systems that will take off and land horizontally like commercial Airplanes

- Using these Scramjet technologies, along with additional ground-and flight-test experiments, will pave the way for affordable and reusable air-breathing hypersonic propulsion systems such as missiles, long range aircraft and space-access vehicles

Advantages:

1. Need not carry oxygen on board
2. No rotating parts makes it easier to manufacture than a turbojet
3. Has a higher specific impulse (change in momentum per unit of propellant) than a rocket engine; could provide between 1000 and 4000 seconds, while a rocket only provides 450 seconds or less
4. Higher speed could mean cheaper access to outer space in the future

Flight Limits/Operating Envelope:



Types of missiles; similarities & differences with launch vehicles;
Control for missiles, airframe components of missiles
Evolution of Space missions: Space missions, mission objectives,
Case studies. Human space flight missions- goals, historical background. The Soviet and US missions. The Mercury, Gemini, Apollo-Soyuz, Space Shuttle
Types of orbits and maneuvers, International Space Station, extravehicular activity. Life support systems.
History and evolution of ISRO; missions carried by ISRO

Missiles:

In military parlance, missiles are powered / guided munitions are broadly categorised as follows:

- A powered, guided munition that travels through the air or space is known as a military **missile** (or *guided missile*.)
- A powered, *un*guided munition is known as a **rocket**.

TECHNOLOGY :

Guided missiles have a number of different system components:

- Targeting and/or guidance
- Flight system
- Engine
- Warhead

Basic categorization

Missiles are generally categorized by their launch platform and intended target. Other kinds of military missiles are Glide- Bombs ,Torpedos etc.

Their basic types are

Conventional guided missiles	Cruise missiles	Ballistic misslies
• Air-to-air missile • Air-to-surface missile • Anti-ballistic missile • Anti-satellite weapon • Anti-ship missile • Anti-submarine missile • Anti-tank guided missile • Land-attack missile • Surface-to-air missile • Surface-to-surface missile • Wire-guided missile	• Hypersonic • Supersonic • Long-range subsonic • Medium-range subsonic • Short-range subsonic	• Tactical ballistic missile • Short-range ballistic missile • Theatre ballistic missile • Medium-range ballistic missile • Intermediate-range ballistic missile • Intercontinental ballistic missile • Submarine-launched ballistic missile • Air-launched ballistic missile

Space Launch Vehicle:

A **launch vehicle** is the rocket we see sitting on the launch pad during countdown.

In spaceflight a **launch vehicle** or **carrier rocket** is a rocket used to **carry a payload from the Earth's surface into outer space**. A **launch system** includes the launch vehicle, the launch pad and other infrastructure.

ROLE :

- It provides the **necessary velocity change** to get a spacecraft into space.
- At lift-off, the launch vehicle blasts almost straight up to gain altitude rapidly and get out of the dense atmosphere which slows it down due to drag. When it gets high enough, it slowly pitches over to gain horizontal velocity.

A launch vehicle consists of a series of smaller rockets that ignite, provide thrust, and then burn out in succession, each one handing off to the next one like runners in a relay race. These smaller rockets are **stages.** In most cases, a launch vehicle uses at least three stages to reach the mission orbit.

Types of Launch Vehicles:

Based on Usage

- Expendable launch vehicles are designed for one-time use. They usually separate from their payload, and may break up during atmospheric reentry.
- Reusable launch vehicles, on the other hand, are designed to be recovered intact and used again for subsequent launches.
- For orbital spaceflights, the Space Shuttle was the only launch vehicle with components which have been used for multiple flights.
- Non-rocket spacelaunch alternatives are at the planning stage.

Launch vehicles are often **characterized by the amount of mass** they can lift into orbit. Launch vehicles are also characterized by the number of stages they employ.Other frequently-reported characteristics of launch vehicles are the nation or space agency responsible for the launch, and the company or consortium that manufactures and launches the vehicle. For example, the European Space Agency is responsible for the Ariane V, and the United Launch Alliance manufactures and launches the Delta IV.

TRAJECTORIES AND ORBITS

A **trajectory** is the path an object follows through space. In getting a spacecraft from the launch pad into space, a launch vehicle follows a carefully-chosen ascent trajectory designed to lift it efficiently out of Earth's atmosphere. We have the following categories of trajectories namely,

A Summary of Parameters for Conic Sections.

Conic Section	a = Semimajor Axis	c = One-half the Distance between Foci	e = Eccentricity
circle	$a > 0$	$c = 0$	$e = 0$
ellipse	$a > 0$	$0 < c < a$	$0 < e < 1$
parabola	$a = \infty$	$c = \infty$	$e = 1$
hyperbola	$a < 0$	$\lvert a \rvert < \lvert c \rvert > 0$	$e > 1$

- The specific mechanical energy for an orbit is given as :

$$E = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$$

- Therefore, for a circle E<0 , parabola E=0 , ellipse E<0 and for a hyperbola E>0 .

Once in space, the spacecraft resides in an **orbit.** An orbit is a fixed "racetrack" on which the spacecraft travels around a planet or other celestial body. When selecting an orbit for a particular satellite mission, we need to know where the spacecraft needs to point its instruments and antennas.

The Orbit. We can think of an orbit as a fixed racetrack in space that the spacecraft drives on. Depending on the mission, this racetrack's size, shape, and orientation will vary.

There are many ways to categorize the orbits either by using the time period or by using the inclination. And in order to classify them we **require six classical orbital elements** and they are namely :

The other types of orbits are :

- **A geostationary orbit** is a circular orbit with a period ofabout 24 hours and inclination of 0°. Geostationary orbits are particularlyuseful for communication satellites because a spacecraft in this orbitappears motionless to an Earth-based observer, such as a fixed ground station for a cable TV company.
- **Geosynchronous orbits** are inclined orbits with a period of about 24 hours.
- A **semi-synchronous orbit** has a period of 12 hours.

- **Sun-synchronous orbits** are retrograde, low-Earth orbits (LEO) typically inclined 95° to 105° and often used for remote-sensing missions because they pass over nearly every point on Earth's surface.
- A **Molniyaorbit** is a semi-synchronous, eccentric orbit used for some specific communication missions.